Sailing into a New Age:

The Effect of the International Maritime Organization's Net-Zero Regulation on the Future of Shipping

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Sailing into a New Age:

The Effect of the International Maritime Organization's Net-Zero Regulation on the Future of Shipping

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Abstract

The International Maritime Organization has passed a new legislation in October of 2025 to penalize large, ocean vessels that exceed greenhouse gas fuel intensity targets set by the organization starting in 2028. This could affect the current situation of routes that a ship takes on an annual basis because costs to operate a ship could rise. In this research, a controlled journey will establish the shortest routes sailed by a ship annually without the consideration of fuel price; while a manipulated journey will establish the evasion route and shortest route sailed in relation to the fuel price ratio of Marine Gas Oil and Very Low Sulphur Oil, annually. The null hypothesis was that the average cost of the controlled journeys will have no significant difference between the average cost of manipulated journeys. Whereas the alternate hypothesis was that the average cost of the controlled journeys will have a significant cost difference between the average cost of the manipulated journeys. In the end, there was no significant difference between the average costs of the journeys, however limitations expressed in this research may be critical for future research of this problem. Therefore, the research is a necessary steppingstone to tackle potential supply chain issue in the future.

Acronyms

ACRONYM	DEFINITION	PAGE
IMO	International Maritime Organization	14
GHG	Greenhouse Gasses	14
MARPOL	International Convention for the Prevention of Pollution from Ships	14
SOx	Sulphur Oxides	14
NOx	Nitrogen Oxides	14
CO2eq	Carbon Dioxide Equivalent	14
ECA	Emission Control Area	14
GFI	Greenhouse Gas Fuel Intensity	15
HFO	Heavy Fuel Oil	15
MGO	Marine Gas Oil	15
VLSFO	Very Low Sulphur Oil	15
LNG	Liquified Natural Gas	15
PM	Particle Matter	16
NPV	Net Present Value	18
IFO380	Intermediate Fuel Oil 380	19
TEU	Twenty-Foot Equivalent Units	19
POL	Port of Loading	22
POD	Port of Destination	22
DWT	Deadweight Tonnage	29
LFO	Light Fuel Oil	29
ULSFO	Ultra Low Sulphur Fuel Oil	30
HSHFO	High Sulphur Heavy Fuel Oil	30
MJ	Megajoules	35
kW	Kilowatts	35
kWh	Kilowatt-hours	35
gCO2/MJ	Grams of Carbon Dioxide per Megajoule	35
BT	Base Target	47
DCT	Direct Compliance Target	47

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1. Introduction

Reducing emissions has become a critical point in all industries that impact the environment in one way or another. Especially the supply chain and logistics of everyday goods from the grocery store, supermarket, retail chains, and so forth; which are heavily reliant on international or regional shipping (World Bank Group, 2025). In the 1960s and 70s, scientists drew concern about the amount of emissions that were building and collecting into the atmosphere. Due to the need to reduce emissions, and after many years of assessments and summits, the International Maritime Organization (IMO) historically took first steps in implementing measures to stop greenhouse gases (GHG) from polluting the atmosphere. The Kyoto Protocol, in 1997, was adapted to the International Convention for the Prevention of Pollution from Ships (MARPOL). This protocol would kickstart the way the maritime sector would treat itself given the volume in its worldwide undertaking (IMOc, 2025); leading to the construction of Annex VI of MARPOL to limit harmful Sulfur Oxide (SOx) (IMOd, 2025) and Nitrogen Oxide (NOx) (IMOe, 2025) emissions. However, until recently, there were no mandatory Carbon Dioxide Equivalent (CO2eq) limits on large, ocean-going, maritime vessels which contribute to 85% of the total CO2 emissions from shipping around the world (IMOa, 2025). Now, the IMO has moved forward to adopt a Net-Zero Regulation for Global Shipping (in October of 2025) and set standards for these ships starting on January 1st, 2027 (IMOa, 2025).

In this research, an interactive model will be developed to simulate hypothetical shipping scenarios that abide by current legislation plus the newest regulation that aims to reduce CO2 emissions set by the IMO. These will include Emission Control Areas (ECAs) established in North America, Hawaii, and the United States Caribbean, the use of bunker fuel that meets current emission standards (after January 1st, 2020, ships use fuel that limits the output to 0.5% of mass of SOx per mass of fuel into the air when sailing outside the ECAs (IMOd, 2025)), and the new Net Zero regulation that was adopted in October of 2025. The purpose of this research is to study the effects that this new regulation will have on the cost of sailing a ship over an annual period in the future and to answer the research question: will ship routing be more inclined to change with the new IMO Net Zero regulation? By utilizing relevant research articles from Fagerholt et al. (2015), Fagerholt and Psaraftis (2025), Tan et al. (2021), and many others, the research conducted in this dissertation builds on the existing knowledge of fuel switching, speed optimization, route optimization, energy consumption, emission factors, and ship evasion. In turn, this research will develop an algorithm (interactive model) that can simulate cost of a ship sailing the shortest route annually (controlled journey) and a ship sailing the evasion route and the shortest route given the fuel price ratio annually (manipulated journey), with the additional parameter of the Net Zero regulation on both journeys. The results, however, showed no significant difference between the prices of each journey, but the data gathered from the hypothetical scenarios is interesting, nonetheless.

2. Literature Review

The IMO has set clear boundaries and regulations for ship operators across the world to reduce the impact global shipping has on the environment. Most notably, the implementation of ECAs on coastal regions in Europe and North America. Recently, legislation has been passed by the IMO which will affect the current situation of competition in international markets regarding maritime shipping. Greenhouse Gas Fuel Intensity (GFI) from vessels sailing around the world will be accounted for and reviewed by the IMO on an annual basis. This accountancy and review will determine if a ship will either be rewarded or penalized for meeting or failing to meet a specific GFI target. In this chapter, the literature review will cover current, relevant research articles on strategies that ships will use in order to compete in international markets with the barriers of ECAs. Fuel switching (Fagerholt et al., 2015; Fagerholt and Psaraftis, 2015; Langella et al., 2016; Abadie et al., 2017; Fan et al., 2020), speed optimization, route optimization, and/or evasion strategies Chen et al. (2018), Ma et al. (2020), Li et al. (2020), Tan et al. (2021), Li et al. (2022) are the most influential topics when it comes to sailing in and out of an ECA. Not forgetting other variables, such as emission factors of fuels and energy consumption of ships. The reason for this research is to build an understanding of how ships will operate around the world in respect to ECAs when leaving and entering prominent ports. Furthermore, this paper is designed to add to the academia of international shipping via the previous strategies in regard to ECAs by incorporating the Net-Zero Regulation of the IMO on the journey of these ships.

2.1. Emission Factors on a Well-to-Wake Basis

Well-to-Wake emissions are the center points of the IMO's new regulation because it accounts for all emissions that fuel creates from the well it was extracted to the wake that it creates from the ship (IMOg, 2025). Comer and Osipova (2021) have developed an article brief that reports on fuels, engine types and emission factors that are developed by the well-to-wake approach from these two variables. In order to establish how the authors came to their conclusions on well-to-wake emissions, they established that well-to-wake emissions are the summation of well-to-tank (upstream) and tank-to-wake (downstream) emissions. The authors focus on fuels like Heavy Fuel Oil (HFO), Very Low Sulfur Fuel Oil (VLSFO), Marine Gas Oil (MGO), and Liquid Natural Gas (LNG). Comer and Osipova (2021) use the GREET (Greenhouse gases, Regulated Emissions, and Energy use in Technologies (US DOE (2025)) model to establish the emissions created from the well-to-tank for all four types of fuel and the ship engines associated with them. The authors then determined that the emission factors, from well-to-wake, for HFO, VLSFO, MGO, and LNG were 3.545, 3.734, 3.782, 3.280 g/g, respectively. Comer and Osipova (2021) then stated that this research can be used in policies that are developed by the IMO and other governing bodies that want to reduce the impact of climate change.

2.2. Energy Consumption of Vessels

Energy consumption is another key factor in the IMO's new regulation when it comes to accounting for GFI. Aijjou et al. (2019) go into depth on the energy and exergy balance of containerships. The authors' research was conducted in light of stricter rules with international shipping and pollution. They sought to analyze where the most energy was conducted on a ship during two voyages so they could improve a ship's energy efficiency. In their research, the authors found that 82% of the energy consumed at sea was due to propulsion of a ship and 70.5% of the energy consumed in port was for powering the ship. Interestingly enough, from the power output by the main engine, 34% of the input energy was wasted to the environment. Therefore, they concluded that for a ship to operate with maximum efficiency, there needs to be an improvement in waste heat recovery solutions.

Alternative energy is becoming increasingly common, especially on ships. Huang et al. (2021) establishes an article that discusses the implementations of sustainable energy like solar, wind, and fuel-cell energy in ship power systems. According to the authors, solar energy with solar panels has become a cheaper form of energy in the past 50 years with solar photovoltaic energy costing less than 0.2 USD/kWh. Wind energy is the most accessible when it comes to shipping with there being an abundance of wind around the world. The cost of wind energy is roughly 0.1 USD/kWh. Finally, fuel-cells are more efficient (90%) and cleaner than direct combustion of fuels with products of reactions only being water and electricity, depending on the fuel used. Huang et al. (2021) have also shown that shore power can greatly decrease the fuel consumption of a ship with the US, UK, and China promoting the use of shore power in their ports. The authors did point out that further research is needed in solar energy to operate safely and efficiently at high temperatures. For fuel cells, further research is needed on the capacity to generate power. All in all, these authors have demonstrated that sustainable energy on ships is becoming a thing of the present, rather than a thing of the future.

2.3. Fuel Switching and Speed

Fagerholt et al. (2015), Fagerholt and Psaraftis (2015), and Li et al. (2020) highlight that fuel switching is the popular preference for ships given its low investment and easy conformity compared to other strategies such as scrubber installation, and complete transition to LNG fuel. In a sense, fuel switching is the change of fuel at a certain point in a maritime journey. In this case, ships will switch from fuel that is compliant with ECA's (e.g. MGO) inside the ECA to cheaper fuel (e.g. HFO) outside the ECA (Fagerholt and Psaraftis, 2015). Scrubbers have a benefit on removing SOx and Particle Matter (PM) from the engine and the exhaust of a ship (IMO, 2019), however there will be higher fuel consumption on the ship due to the energy input needed to operate these scrubbers (Abadie et al., 2017). LNG is one the cleanest fuel sources for the marine vessel to use, yet it poses as one of the costliest for ships to implement into their systems due to the high price to buy the fuel and the energy input to maintain the fuel at its

liquified state (Brun and Rørvik, 2023). Given that both approaches involve premium dues from ships, fuel switching has been the prominent plan for marine vessels.

Langella et al. (2016) produce a case study that goes into detail on the impact of fuel switching and the effects of emissions from ships (SOx, NOx, and Particle Matter (PM)) have on the surrounding area while taking three scenarios into account. The first scenario is the Heavy Fuel Oil (HFO) scenario. This scenario has the ship use HFO for the entire duration of the journey moving into port. The second scenario is the fuel switching scenario. This involves switching from Heavy Fuel Oil to Marine Gas Oil while moving into port. The third scenario is the Marine Gas Oil (MGO) scenario. This scenario involves MGO being used for the entire duration of ships moving into port. The model that the authors used was the Gaussian Model ISC which is able to calculate the concentration of pollution from ships and the air quality from that pollution. The first scenario illustrates a huge concentration of pollution in the surrounding area of the port with some concentration between 8-10 µg/m^3 of pollutants. The second scenario has less concentration of pollution (7-9 µg/m^3 of pollutants) and even more so in the third scenario (6-7 µg/m^3 of pollutants). The authors then concluded that fuel switching has a substantial benefit of reducing emissions.

The fuel switching strategy needs to be explained further in order to grasp a basic understanding of this paper's research. Fagerholt et al. (2015) have written in their article about optimizing routes and speed inside and outside of an ECA using a mathematical optimization model on the basis of fuel switching which can help mitigate fuel cost. Rather than consuming expensive MGO for the duration of the ship's journey to comply with ECA regulations, ships will switch from MGO to HFO once they have left the ECA. Furthermore, rather than sailing directly into an ECA, their study has shown that ships will try to avoid the area as much as possible, even by taking the longer route from port to port by reducing time spent in an ECA. Inevitably, ships do need to sail inside an ECA. The authors declared that ships will most likely cruise at lower speeds consuming MGO inside the area and switch to HFO with higher speeds outside the area. With this increase in distance sailed and speed accumulated, they declared that the fuel switching strategy will reduce fuel cost. Furthermore, they stated that the price difference between their two tested fuels (MGO and HFO) would determine the route in which ships would take in order to cut costs. To conclude, the authors have inclined that even though ECAs were implemented to reduce emissions such as SOx, more fuel would be consumed from these longer routes. Moreover, switching to fuels like HFO and increasing the speed of a ship would ultimately cause more emissions such as CO2 and SOx to be emitted into the air. Sequentially, the authors left the window open to further research on the true effectiveness of an ECA and the possibility of implementing a stricter ECA policy.

This segues into the next literature Fagerholt and Psaraftis (2015) which is the extension of Fagerholt et al. (2015) but concentrates on speed optimization issues with ships that comply with ECA regulations by fuel switching. The defining problem presented in this study is the maximization of profits of ships. In this article, routes that were given had one port inside of an

ECA and the other outside the ECA. The key factor in their problem was a refraction point. The authors define a refraction point as the location where the ship leaves an ECA. So, the refraction point had to be the shortest distance away from the port inside an ECA. Additionally, they draw attention to their own literature review of Ronen (1982) who developed a mathematical model for income generation of one leg of a journey of a ship. The problem that Ronen (1982) faced was the optimal sailing speed of a ship given that the function for fuel consumption is cubic. As speed increases, fuel consumption over time increases by a power of three. Having this knowledge, they used an extension of Ronen's (1982) model to solve their own problem with an ECA. This problem being optimal speed inside and outside of an ECA. Fagerholt and Psaraftis (2015) talk about three sensitivities. Firstly, the fuel sensitivity which clarifies that the ratio between the MGO and HFO increases the cost of fuel does as well. On the other hand, when the ratio decreases, so does the cost of fuel. Most importantly, when the price of fuel increases, the speed at which a ship sails inside an ECA decreases. Secondly, the authors discuss the revenue sensitivity which clarifies the ratio of revenue to the speed of the vessel. If the revenue ratio increases, the speed of the ship increases as well. This is due to the performance of the market. If the market is good, ships have a higher incentive to get to destination quicker; if the market is bad, the ship has a lower incentive to get to destination quicker. Thirdly and most relevant to Fagerholt et al. (2015) further research, Fagerholt and Psaraftis (2015) discuss the distance sensitivity clarifying the ratio of distance from the port to ECA boundary and the total distances of the voyage. As the distance of the port to ECA boundary increases, the speed in which the ship sails decrease. In the end, Fagerholt and Psaraftis (2015) wanted their paper to be a utility for route optimization when the route encounters an ECA.

Nevertheless, it must be remembered that more fuel consumed from longer routes and higher speeds using the fuel switching strategy means more CO2 and SOx emissions released into the air (Fagerholt et al., 2015). Therefore, scrubbers need to be studied as well, in order to be certain that fuel switching is the correct strategy to use. Abadie et al. (2017) have written an article about the use of alternative fuel compared with scrubbers on a ship to reduce emissions. They use a stochastic model for the unknown price of bunker fuel, which is claimed by the authors to be strongly correlated to the crude oil prices. When it comes to scrubbers, the authors determined that the remaining lifetime of the ship and how long the ship spends in an ECA are deciding factors. The older the ship, the greater the value of installing a scrubber rather than switching fuel is. The longer the ship spends inside an ECA, the greater the value of installing a scrubber because the ship can continue to use cheaper fuel rather than switch to more expensive fuel. Abadie et al. (2017) noted that their research can be used to assess long-term investments, but they call for further research using their paper on policies to regulate CO2 and other emissions.

Another article by Fan et al. (2020) tests the comparison of using scrubber's vs fuel switching. The authors use net present value (NPV) as a cost-benefit analysis between the two strategies. For those ships that travel transoceanic, it is beneficial to implement the fuel switching

strategy because the ECAs are only a small percentage of their journey and therefore don't need to invest in fuel scrubbers. On the other hand, regional shipping, especially those ships that sail the majority of their journey in ECAs, are more prone to adapt their vessels with scrubbers because of the little (or none) time they spend outside the ECA. Nonetheless, there is a dependency on fuel price that affects the choice in which ships can take in order to comply with regulations. Again, there is a gap for future research in this study. The authors only focused on the sulfur regulations in their study, so there is a call to further investigate regulations of other emissions.

2.4. Route Optimization and Evasion Strategy

Compliance with ECA regulations is unavoidable for ships; therefore, many will try to avoid the ECA for as long as possible, Fagerholt et al. (2015). For this reason, Li et al. (2020) wrote an article on optimal sailing and evasion strategy selection. The work that they did dealt with fuel switching of Intermediate Fuel Oil (IFO380) and MGO. They specifically focus on price ratio when the decision to evade or to not evade an ECA presented itself on a specific path from two fixed ports. Additionally, they use two mixed integers nonlinear programming models: one to calculate the ship's cost and the other to create a threshold for the fuel price ratio. More importantly, they found that the longer the distance to the destination port and the greater the fuel price ratio, the more cost saving for the ship. According to the authors, after 2020, the evasion of ECAs will still be an inviting strategy to execute with 29.05% of ships still adopting it on their routes. Since this attractive strategy is still going to be used, the ultimate problem that the authors concluded was an evasion strategy leads to an increase in pollution due to the distance travelled and must not be ignored.

Chen et al. (2018) show that the size of a ship can impact the amount of CO2 and SOx being emitted. In their study, they conducted their investigation on ships coming from Asia to Europe with the hypothetical introduction of an ECA into the Mediterranean Sea. They analyzed the routes taken from Asia to Europe. The Suez Canal is the gateway for ships coming from Asia to Europe, and the hypothetical introduction of an ECA in the Mediterranean Sea will affect the probability of ships choosing this route. As ship size increases up to 5000 Twenty-Foot Equivalent Units (TEUs) the probability for ships to choose this route increases. However, once the size goes past 5000 TEUs, the probability to choose this route decreases significantly. They also found that if a ship size increases, emissions will decrease to a certain degree. Smaller ships, below 5000 TEUs, will reduce a significant amount of emission, but the impact of large containerships is quite moderate. The authors then conclude their article with a call for further research on the studies of international policies on emission of shipping.

Ma et al. (2020) reveal the route optimization inside and outside an ECA via the implementation of a cell-based method while including regulations of an ECA and weather conditions. They discuss the three routes that are presented which are the Shortest Route, Weather Route, and then the Fuel Cost Route. The model used in this article is an integer

programming model calculating all routes that are possible and their costs. This is compared with three other methods (e.g. Dijkstra, A-Star, and ES-isochrone) and the actual route taken by the ship. The Dijkstra method is used to calculate the short distance from one vertex (or node) to all the other vertices (nodes). In this case, Dijkstra method was used to create a route to the maritime port based on given parameters. A-Star is the improvement of Dijkstra algorithm where it takes problem solving capabilities to find the end node from the start node more efficiently. It is quite easy how they accomplished their proposed model to optimize the best route for a ship. First, they used Dijkstra algorithm to narrow down optimized routes. Then they used CPLEX from IBM to solve their integer programming model. The research showed that emissions inside an ECA can be reduced significantly from the route optimization, but there would likely be an increase in emissions outside of the ECA. This is due to the increase in total sailing distance. The authors found in their study that ECA policies might be strengthening shipping operators' efforts to avoid ECAs. Therefore, international pollution might increase, so the authors call for further research on their study with cleaner fuels.

Tan et al. (2021) focuses their study on the cost of evasion strategy and the non-evasion strategy of an ECA. Given the new ECA boundary in China, their research was on two ports inside an ECA with a capricious coastline, which reflect that of Chinese ports and the ECA. The model that was chosen was a geometric model to measure the route that a ship would sail using the evasion strategy. Even more interesting, they simulated distance increase of the ECA line to the coastline. As the distance of the ECA line increased, the use of the evasion strategy increased to a point and then decreased as the distance increased. Finally, the authors found that future research on creating a new model would show the impact of ECAs from different countries.

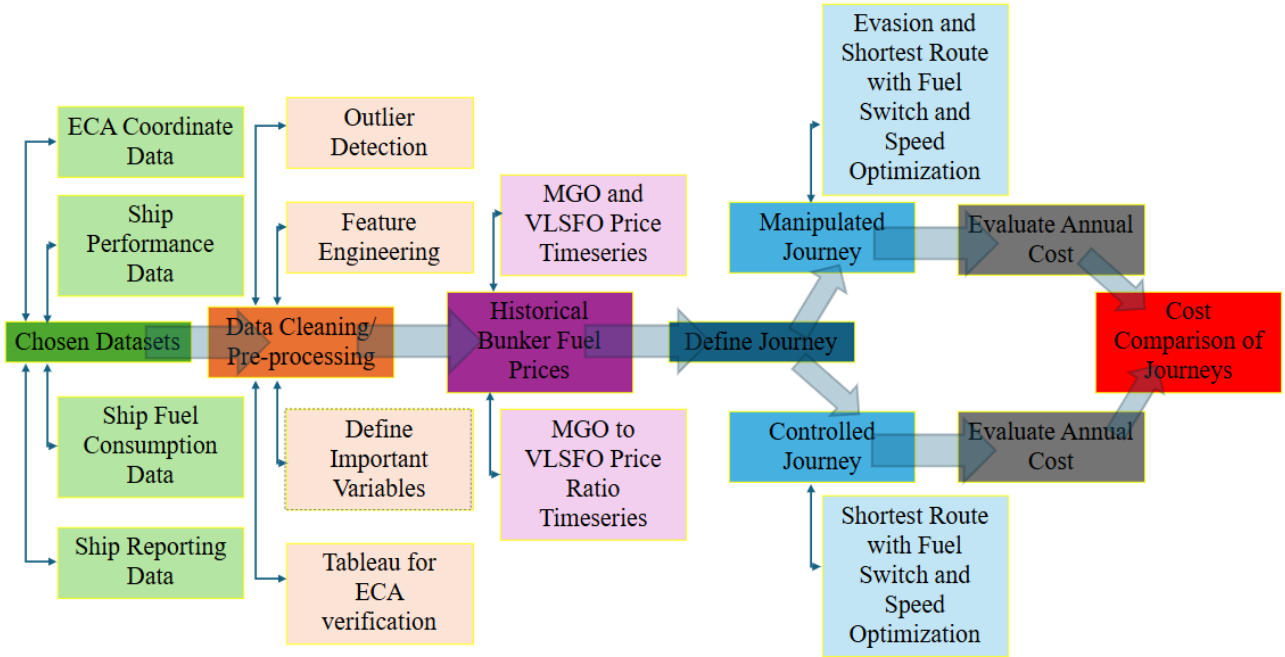
Li et al. (2022) wrote an important piece of literature because they take into account the evasion strategy of ships into the assessment of emission reductions with the implementation of an extreme ECA boundary. This article is an extension to their previous research in Li et al. (2020) but in this case, they focus more on the ECA boundary to assess the effectiveness that it has on maritime shipping. In a sense, their research is conducted to design an ECA where it makes it impossible to take an evasion strategic route. Influencing factors that the authors found in their study were: coastlines, fuel price, and ship type. Coastlines impacted how extreme ECA could be implemented because the authors refer to the coastline and the port to determine a rational boundary. Fuel price has an impact on an extreme ECA boundary because if the ratio of fuel price increased, which made the incentive to evade the ECA. Ship type has little impact on extreme ECA boundaries because it has more of an impact on fuel cost. Ultimately, the authors found that the implementation of extreme ECA boundaries increased cost of fuel but decreased emissions from ships as well as preventing ships from taking an evasion strategic path. Nonetheless, the fact is that extreme ECAs and implemented ECAs have the same effect of increasing fuel cost and decreasing emissions. The significant difference is the amount of increase and decrease of fuel cost and emissions, respectively. The future research that was asked by the authors was the improvement of fuel price forecasting and fuel consumption models.

In conclusion, research on fuel switching, route optimization, speed optimization, and evasion strategies have been in abundance. Fuel switching has a substantial benefit to reduce emissions (Langella et al., 2016) and is the most popular strategy for ship operators that encounter an ECA on their journey (Fagerholt et al., 2015; Fagerholt and Psaraftis, 2015; Li et al., 2020). Furthermore, route and speed optimization inside and outside an ECA has become crucial for ship operators to stay competitive in international markets (Ma et al., 2020; Chen et al., 2018; Fagerholt and Psaraftis, 2015). Even more so, evasion strategies have now become the new normal for ship operators (Li et al., 2020) which have brought into discussion about stricter ECAs (Li et al., 2022; Tan et al., 2021, Fagerholt et al., 2015). However, there are gaps in the research that need to be addressed. Fagerholt et al. (2015) highlighted that there is a need for stricter ECA policies and Fagerholt and Psaraftis (2015) want their paper to be used as a guide to optimize shipping routes when the routes run into ECAs. Li et al. (2020) have stated that given the increase in total distance travelled from evasion strategies by ship operators, pollution will be higher and can't be ignored. Ma et al. (2020) said that due to the increase in pollution, cleaner fuels must be explored. Tan et al. (2021) said there needs to be a new model to illustrate the impact of an ECA in different countries. Li et al. (2022) were adamant to the improvement of fuel price forecasting and fuel consumption models. Finally, Abadie et al. (2017), Chen et al. (2018) and Fan et al. (2020) say that further research is needed for policies to regulate CO2 and other pollution emissions. Now, there is justification for this paper to add to the academia of international shipping via fuel switching, route optimization, speed optimization, and evasion strategies regarding ECAs by incorporating the Net-Zero Regulation of the IMO on the journey of these ships.

3. Methodology

This section defines the steps taken to compare the cost of evasion and shortest routes (Manipulated Journey), and the cost of the shortest route only (Controlled Journey), for a containership that sails between two ports and encounters two ECAs on an annual basis. The Manipulated Journey is constructed to emulate cost-minded shipping throughout the year. Given that fuel consumption costs are between 30-60% of the operating cost of a ship (Yıldız, 2025), the trigger to change from evasion route to shortest route (and vice versa) will be dependent on the fuel price ratio between MGO and Very Low Sulfur Fuel Oil (VLSFO). To supplement the ratio, historical bunker fuel prices will be used to simulate real fuel pricing scenarios for ships going forward. The Manipulated Journey will incorporate fuel switching; route optimization inside an ECA; speed optimization inside and outside an ECA; and evasion of a ship when the port of destination (POD) is inside an ECA. The Controlled Journey is constructed to emulate the fastest shipping between the port of loading (POL) and the POD so that containerships can maximize their delivery of containers in a year. Furthermore, the Controlled Journey will only incorporate fuel switching and speed optimization inside and outside and ECA. Throughout the durations of these journeys, emissions and energy production of a ship will be monitored and collected, so that by the end of the year, ships can calculate the annual GFI attained. GFI's that are above a certain target are subject to remedial units purchased from the IMO, therefore adding to the total cost to operate a container ship. By calculating the total cost accrued over a year, ship operators can more effectively evaluate the future routing of ships when taking into account the new IMO Net Zero Regulation which monitors GFI of a ship on a well-to-wake basis.

Figure 1: Step process of choosing data to evaluation of journeys



Source: Author

3.1. Chosen Datasets: Data Sources, Collections, and Cleaning

3.1.1. “ECA_NA_P”, “ECA_NA_A”, “ECA_NA_H”, “ECA_US_C” Datasets

ECA coordinates are the most important pieces of data for this research. The ECA boundary location is vital to establish the percentage of distance spent inside and outside an ECA in order to correctly identify optimized ship speeds in both journeys. The coordinate data for these ECAs [1] and implemented into Tableau. However, when implementing the coordinates into Tableau, the platform can only understand Decimal Degrees (e.g. 0.00N,0.00W). On the IMO website, coordinates are listed in Degree-Minute-Seconds (e.g. 0.00°, 0.00’, 0.00”). Therefore, they must be changed to suit the platform. After copying and pasting coordinates from the website into an excel spreadsheet, parsing text to columns is the next step. This will be to parse away the symbols that are not needed (°, ‘, ”), leaving only the numbers in respective columns defined as LAT D, LAT M, and LAT S (as well as LON D, LON M, LON S). Following the parsing, the addition of all three numbers into a separate respective column defined as Latitude (and Longitude) will be done as such:

$$\text{LAT D} + (\text{LAT M} * 60) + (\text{LAT S} * 3600)$$

$$-1 * (\text{LON D} + (\text{LON M} * 60) + (\text{LON S} * 3600))$$

(Note: Any Longitude Coordinates to the west of the Prime Meridian are negative)

This will give the correct decimal degree numbers needed to implement into Tableau for the visualization of the coordinates on a map. Still, there needs to be cleansing of the LAT/LON D, LAT/LON M, and LAT/LON S from the dataset in order to only leave behind the Latitude and Longitude columns. This is where Python programming language will be utilized with the titled code: ECA_Coordinate_Finder. By importing the excel files (saved as “ECA North America Pacific.xlsx”, “ECA North America Atlantic.xlsx”, “ECA North America Hawaii.xlsx”, and “ECA US Caribbean.xlsx”) into Spyder Integrated Development Environment (IDE) for python, useless columns will be dropped leaving only “Latitude”, “Longitude” and “Point” columns (“Point” column is a counter for how many coordinates there are).

[1] The coordinates can be found via https://www.imo.org/en/ourwork/environment/pages/emission-control-areas-(ecas)-designated-under-regulation-13-of-marpol-annex-vi-(nox-emission-control).aspx

Figure 2: Example output of dataset in the ECA_Coordinate_Finder.py code

	POINT	LATITUDE	LONGITUDE
0	1	22.548333	-153.009167
1	2	23.101389	-153.476667
2	3	23.536389	-154.036667
3	4	23.863056	-154.613333
4	5	24.363611	-155.853611

Source: Author

The following figures illustrate the coordinates of the North American Pacific, North American Atlantic, Hawaii and US Caribbean ECAs on a map in the visual analytics platform, Tableau.

Figure 3: ECA coordinates for North America Pacific

Source: Author

Figure 4: ECA coordinates for North America Atlantic

Source: Author

Figure 5: ECA coordinates for North America Hawaii

Source: Author

Figure 6: ECA coordinates for United States Caribbean

Source: Author

3.1.2. “ship_performance” Dataset

Secondly, the “Ship Performance Clustering” dataset, made by Fijabi J. Adekunle, was downloaded from Kaggle[2] website to acquire data on ships and shipping routes. The setting where this data was collected was from the Gulf of Guinea. In Python, Google Colab, the dataset will be imported and displayed to help understand what data is needed and not needed.

[2] The data can be found via https://www.kaggle.com/datasets/jeleeladekunlefijabi/ship-performance-clustering-dataset

Figure 7: Ship Performance Clustering Dataset before cleaning

	Date	Ship_Type	Route_Type	Engine_Type	Maintenance_Status	Speed_Over_Ground_knots	Engine_Power_kW	Distance_Traveled_nm	Draft_meters	Weather_Condition	Cargo_Weight_tons
0	2023-06-04	Container Ship	NaN	Heavy Fuel Oil (HFO)	Critical	12.597558	2062.983982	1030.943616	14.132284	Moderate	1959.017882
1	2023-06-11	Fish Carrier	Short-haul	Steam Turbine	Good	10.387580	1796.057415	1060.486382	14.653083	Rough	162.394712
2	2023-06-18	Container Ship	Long-haul	Diesel	Fair	20.749747	1648.556685	658.874144	7.199261	Moderate	178.040917
3	2023-06-25	Bulk Carrier	Transoceanic	Steam Turbine	Fair	21.055102	915.261795	1126.822519	11.789063	Moderate	1737.385346
4	2023-07-02	Fish Carrier	Transoceanic	Diesel	Fair	13.742777	1089.721803	1445.281159	9.727833	Moderate	260.595103

Operational_Cost_USD	Revenue_per_Voyage_USD	Turnaround_Time_hours	Efficiency_nm_per_kWh	Seasonal_Impact_Score	Weekly_Voyage_Count	Average_Load_Percentage
483832.354540	292183.273104	25.867077	1.455179	1.415653	1	93.769249
483388.000509	883765.787360	63.248196	0.290361	0.885648	6	93.895372
448543.404044	394018.746904	49.418150	0.499595	1.405813	9	96.218244
261349.605449	87551.375175	22.409110	0.702906	1.370704	1	66.193698
287718.375160	676121.459632	64.158231	1.331343	0.583383	8	80.008581

Source: Author

In this research, the type of ship that will be focused on is Container Ships with a long-haul route because the conducted research deals with container ships and long-haul routes from ports in the continental United States to ports in the Caribbean and Hawaii. Therefore, there is no need to deal with vessels in the data like for ships (e.g. Fish Carriers, Bulk Carriers, and Tankers) and route types (e.g. short-haul, trans-oceanic, and coastal). This brings the total number of 2736 rows of data to 173 rows utilized in the dataset. Data which are not utilized in the dataset were: engine type, maintenance status, draft meters, weather conditions, revenue per voyage, efficiency of nautical miles per kilowatt hour, seasonal impact score, and weekly voyage count. The reason for not utilizing these types of data was because they did not contribute to the goal of the research, which is to compare the cost of aforementioned journeys. Crucial data, such as speed in knots and operational cost, will be used in implementing into speed optimization equations for the ship sailing inside and outside an ECA, like maximum speed and average operation cost of a containership travelling long-haul journeys.

3.1.3. “ship_fuel_consumption” Dataset

The third dataset is from the IMO’s Data Collection System[3] on the 2023 reporting period of fuel oil consumption data and number of ships that submitted the data. Again, this will be crucial for the implementation of fuel consumption in speed

[3] The data can be found via https://wwwcdn.imo.org/localresources/en/OurWork/Environment/Documents/Reporting%20year%202023.pdf

optimization equations. The IMO's Data Collection System on the 2023 reporting period of fuel oil consumption data was used as a two-part dataset in this research. The first part is the fuel consumption of Container Ship ranging from less than 10,000 DWT to 200,000 DWT and above, and the inclusion of total consumption of ships based on Dead Weight Tonnage (DWT). Notably, the types of fuel consumed by these ships like MGO, HFO, Light Fuel Oil (LFO), and Liquified Natural Gas (LNG).

Figure 8: The 2023 reporting period of fuel oil consumption for Containerships before cleaning

	Column1	Diesel /\nGas Oil\n(MDO /\nMGO)	Heavy Fuel\nOil (HFO)	Light Fuel\nOil (LFO)	Ethane	Ethanol	Liquefied\nNatural\nGas (LNG)	Liquefied\nPetroleum\nGas (LPG)\n- Butane	Liquefied\nPetroleum\nGas (LPG)\n- Propane	Methanol	Other
0	Containership	4858186.0	43113643.0	10182858.0	0.0	0.0	468540.0	20064.0	3990.0	0.0	310050.0
1	Less than 10,000 DWT	179784.0	465430.0	159148.0	0.0	0.0	0.0	0.0	0.0	0.0	138.0
2	10,000 ? DWT < 15,000	387670.0	1668478.0	573355.0	0.0	0.0	3467.0	0.0	0.0	0.0	1417.0
3	15,000 ? DWT < 40,000	848742.0	7948001.0	2347012.0	0.0	0.0	24400.0	0.0	3990.0	0.0	6940.0
4	40,000 ? DWT < 80,000	1288696.0	10344054.0	3161549.0	0.0	0.0	13898.0	0.0	0.0	0.0	73723.0
5	80,000 ? DWT < 120,000	1041756.0	9777012.0	1855205.0	0.0	0.0	5326.0	0.0	0.0	0.0	87810.0
6	120,000 ? DWT <\n200,000	978610.0	10415009.0	1832604.0	0.0	0.0	271426.0	0.0	0.0	0.0	137286.0

Source: Author

3.1.4. "ship_reporting" Dataset

The second part of the dataset is the number of ships and their types (e.g. Container ships) that submitted these reports to the IMO. The same ranges apply to this dataset as the rows are identical to the previous figure (Figure 8). Most notably in this dataset, the number of ships that reported in their fuel consumption based on their DWT to the IMO in the year 2023.

Figure 9: The 2023 reporting period of number of ships for Containerships before cleaning

	Column1	Number\nof ships	Gross tonnage	Deadweight\ntonnage	Distance travelled	Hours under\nway	CO_{2}\nemissions	AER for\neach\nEEDI\nship Size	cgDIST\nfor each\nEEDI\nship Size
0	Containership	5076.0	279735936.0	317809559.0	371674845.0	28556513.0	183739286.0	7.31	-
1	Less than 10,000 DWT	242.0	1675570.0	2043832.0	12773831.0	1154197.0	2527550.0	23.31	-
2	10,000 ? DWT < 15,000	616.0	6169096.0	7763370.0	36815156.0	3104221.0	8258176.0	17.82	-
3	15,000 ? DWT < 40,000	1634.0	36141260.0	45221966.0	110166487.0	8731001.0	34956815.0	11.34	-
4	40,000 ? DWT < 80,000	1122.0	54350135.0	64631189.0	90747365.0	6829180.0	46393513.0	8.81	-
5	80,000 ? DWT < 120,000	668.0	59680995.0	67352581.0	56778551.0	4082967.0	39764038.0	6.90	-
6	120,000 ? DWT < 200,000	654.0	92250986.0	98751618.0	54580390.0	3945974.0	42365386.0	5.15	-

Source: Author

3.1.5 “ship_fuel_consumption_reporting” Dataset

The two datasets will then be merged in Google Colab on the ‘Column1’ column. This merged dataset will be called “Ship_Fuel_Consumption_Reporting”. The data that will not be utilized were: Heavy Fuel Oil, Ethane, Ethanol, Liquified Natural Gas, Liquid Petroleum Gas – Butane, Liquid Petroleum Gas – Propane, Methanol, Other, Distance travelled, Hours under way, CO2 emissions, AER for each ship size, cgDIST for each ship size, Gross Tonnage, and Deadweight tonnage. The reason for the data not being utilized is because of their unimportance to this research. However, it is noticed that CO2 emissions are recorded in the dataset, which is something that is important in the research. CO2 emissions collected in this dataset are not defined to be emissions from a well-to-wake basis, so it will be dropped from the dataset. Note that LFO is kept in the dataset. LFO consumption will be assumed to be fuel consumption of VLSFO because LFO is associated with Ultra Low Sulfur Fuel Oils (ULSFOs) and VLSFOs according to (IMOf, 2025). While HFO is also associated with VLSFO, it is associated with High Sulfur Heavy Fuel Oil (HSHFO) as well. Therefore, the assumption to consider VLSFO in the category of LFO is important in the realm of emissions.

Figure 10: Combined datasets for the 2023 reporting period after cleaning

	Column1	Diesel /\nGas Oil\n(MDO /\nMGO)	Light Fuel\nOil (LFO)	Number\nof ships
0	Containership	4858186.0	10182858.0	5076.0
1	Less than 10,000 DWT	179784.0	159148.0	242.0
2	10,000 ? DWT < 15,000	387670.0	573355.0	616.0
3	15,000 ? DWT < 40,000	848742.0	2347012.0	1634.0
4	40,000 ? DWT < 80,000	1288696.0	3161549.0	1122.0
5	80,000 ? DWT < 120,000	1041756.0	1855205.0	668.0
6	200,000 DWT and above	132894.0	253985.0	139.0

Source: Author

3.1.6. “bunker_fuel_prices” Dataset

The final dataset is from the United States Department of Agriculture[4] website on historical bunker fuel pricing data and will be crucial for simulating prices of fuel throughout a given year. The average prices for MGO and VLSFO across 20 major ports around the world are collected and stored in this dataset. This data will help simulate the price increase and price decrease of bunker fuel in future years by using the most recent year, 2024. The reason for this year, in particular, is because it is the most recent full year of data that closely emulates what future data of fuel prices might be. Moreover, this data is crucial to the total cost of operating a ship on an annual basis. Data that was not used was Intermediate Fuel Oil 180cSt and Intermediate Fuel Oil 380cSt because they have no relevance to the research being conducted.

Figure 11: Historical bunker fuel prices for the year 2024

	Date	Month	Year	VLSFO Fuel Oil, IMO 2020 Grade, 0.5%	Marine Gas Oil
1283	2024-01-01	January	2024	637.0	854.0
1284	2024-01-02	January	2024	636.0	852.0
1285	2024-01-03	January	2024	633.5	840.0
1286	2024-01-04	January	2024	633.5	841.5
1287	2024-01-05	January	2024	635.0	841.0

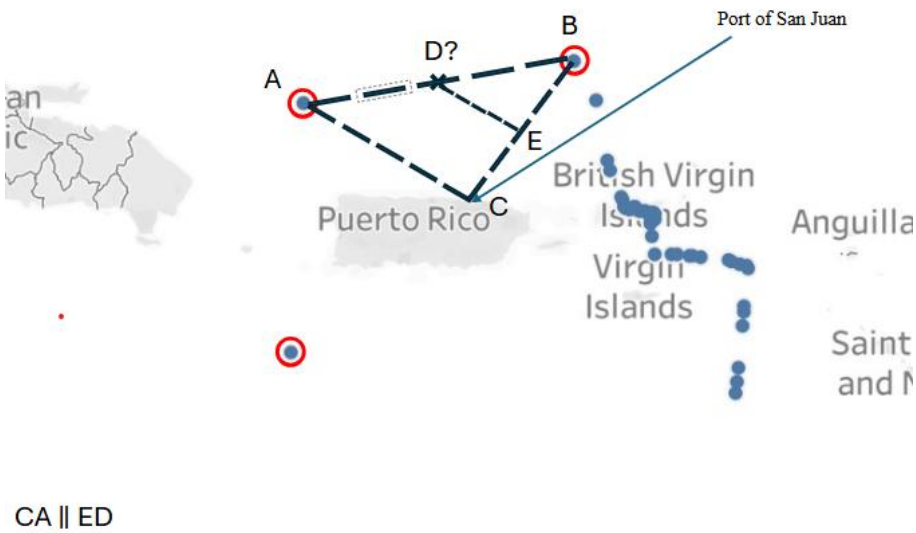
Source: Author

3.2. Data Pre-Processing: Outliers, Feature Engineering, and Variables

The coordinate data only needs one step of preprocessing before executing the building phase of the research. Notice in the ECA US Caribbean there are only three coordinate points to establish the north and west sides of the ECA:

[4] Data for bunker fuel prices can be found via https://agtransport.usda.gov/Fuel/Daily-Bunker-Fuel-Prices/4v3x-mj86/about_data

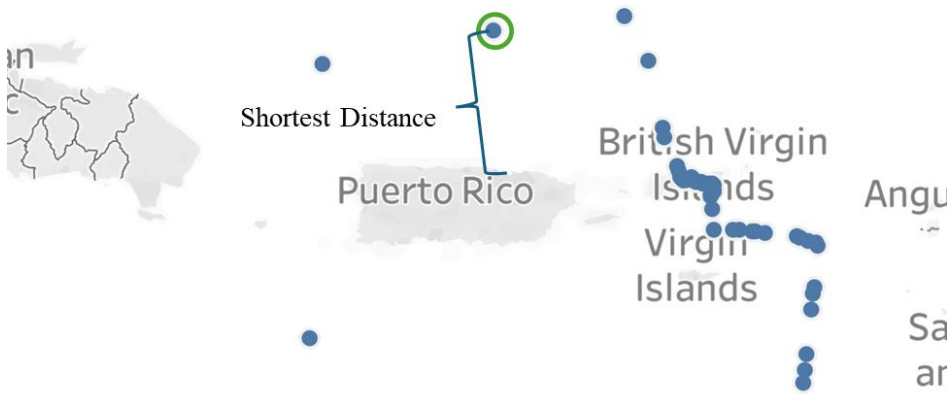
Figure 12: ECA US Caribbean missing crucial coordinate point



Source: Author

This can present a problem when routing evasion routes and shortest routes from POL to POD. Given that the US Atlantic Seaboard is to the Northwest of the port of San Juan (A containership port in the Caribbean), there can be an inaccuracy when establishing the evasion route distance to the ECA boundary. If there were not an added coordinate point in this scenario, then the shortest route might have a smaller distance percentage inside an ECA than that of the evasion route. This is not what can happen. The evasion route must have a smaller distance percentage inside an ECA than the shortest route, or else it would render the evasion route insignificant. To fix this, another coordinate point was added to ECA_US_C dataset using the midpoint theorem. Where point E and point D are the respective midpoints of line CB and line BA. This result gives the following figure:

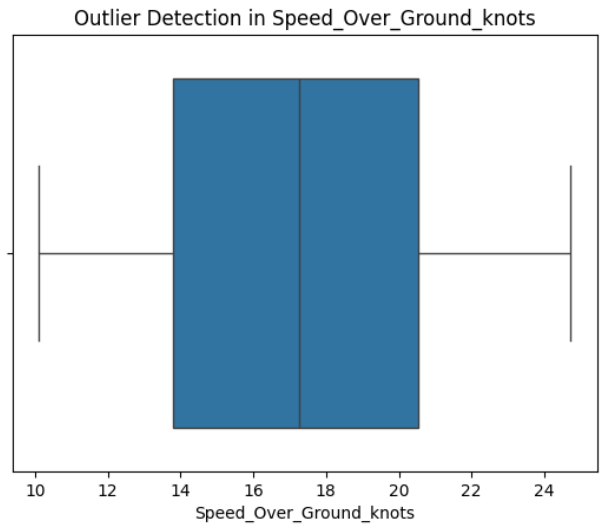
Figure 13: ECA US Caribbean with crucial coordinate point

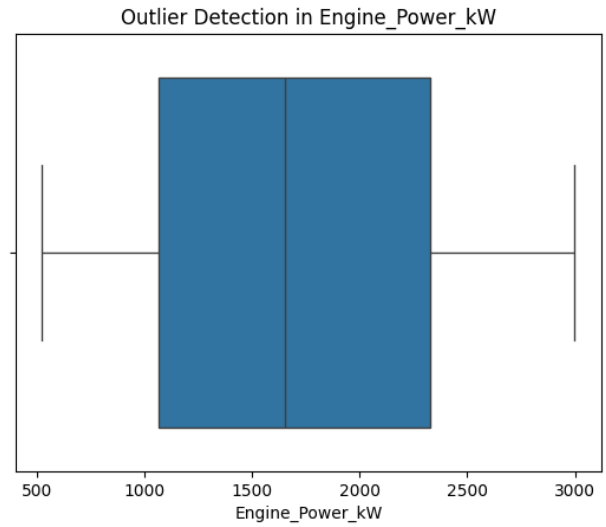


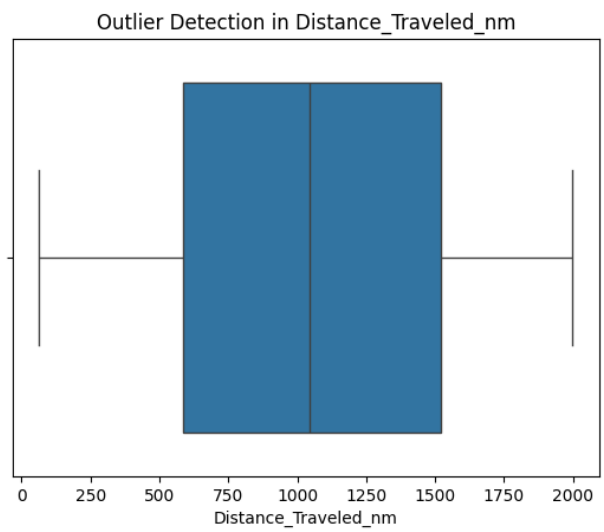
Source: Author

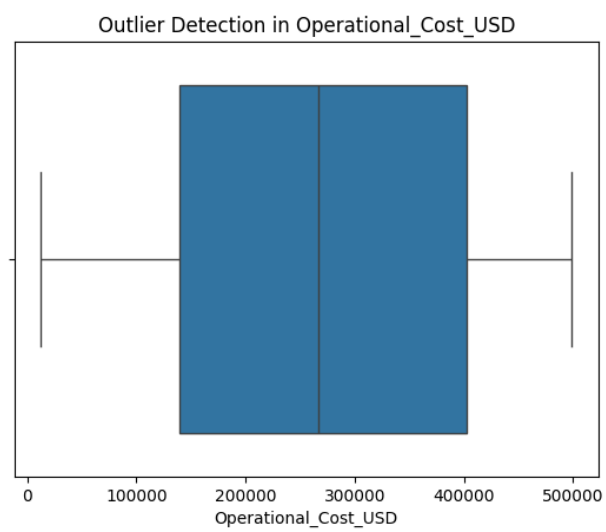
Next, container_performance dataset needed outlier detection in order for the data quantities to not skew the results in any way. Outlier detection was done to columns of speed_over_ground_knots, engine_power_kW, distance_travelled_nm, operational_cost_USD, turnaround_time_hours, and average_load_percentage. The reason for the outlier detection on these columns is because of their impact on the results of this research. If data were skewed, it would not reflect reality correctly.

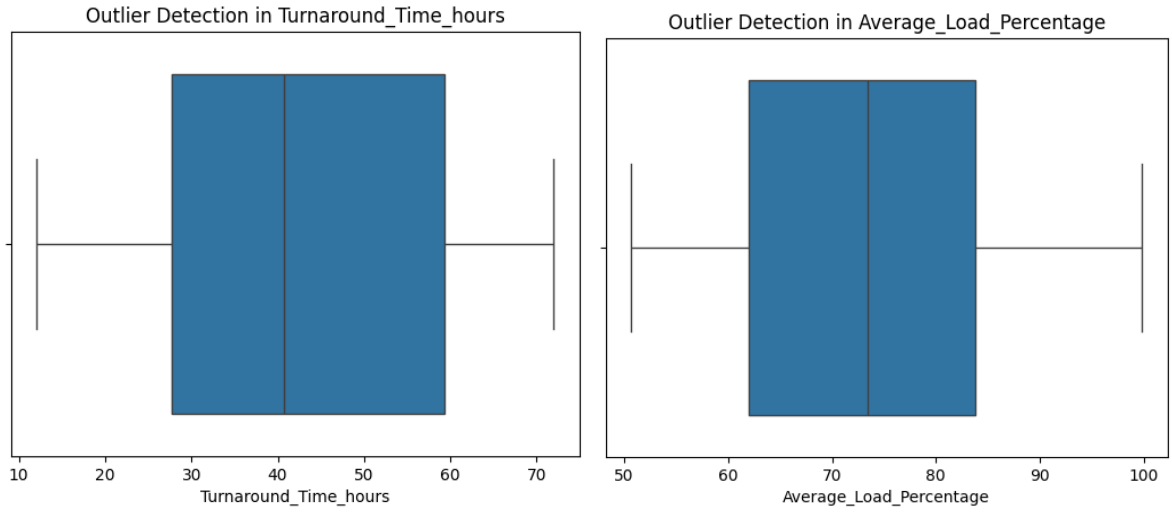
Figures 14, 15, 16, 17, 18, 19: Outlier detection of key columns in conatiner_performance dataset











Source: Author

Following the outlier detection, which resulted in no outliers, container_performance dataset will have two feature engineering criteria added. These are Time of the Journey (hrs) and Power Consumption to Time (MJ). Both variables go hand-in-hand with each other. The resulting GFI will be in units of gCO2/MJ, therefore solving for Time of the Journey in hours of a ship will be one part of solving power consumption to time of a ship. Solving the Time of the Journey involves dividing the distance of a ship by the ship's speed. After time has been solved, Power consumption to time will involve multiplying engine power (kW) by the Time of the Journey of a ship. Although this will give the resulting units as kilowatt-hours (kWh), thus using the conversion factor for kWh to Megajoules (MJ), the resulting product will be power consumption to time in MJ (1 MJ = 3.6 kWh). After these two features were added, the dataset was checked for duplicates and NAN values, resulting in no duplicates or NAN values.

Figure 20: Detection for duplicates and NAN values in container_performance dataset

```
NAN values in the Container Performance dataframe:
 Date                              0
Ship_Type                          0
Route_Type                         0
Speed_Over_Ground_knots            0
Engine_Power_kW                    0
Distance_Traveled_nm               0
Cargo_Weight_tons                  0
Operational_Cost_USD               0
Turnaround_Time_hours              0
Average_Load_Percentage            0
Time_of_journey(hrs)               0
Power_Consumption_to_Time(MJ)      0
dtype: int64

Number of Duplicates in the Container Performance dataframe:  0
```

Source: Author

Figure 21: Feature engineering of container_performance dataset

Time_of_journey(hrs)	Power_Consumption_to_Time(MJ)
31.753359	188449.961128
68.402523	661277.696319
96.697491	642530.935735
107.194159	441532.472785
39.097752	337507.814355

Source: Author

The ship_fuel_consumption_reporting dataset requires two features engineered into it as well. Both of which will be prepared to be in the speed optimization equation. The first is MGO fuel consumption. By dividing the number of ships by the annual fuel consumption of MGO, the research can have an accurate account of MGO fuel consumption per ship by tonnage a day.

However, given the speed optimization equation in this research utilizes liters per hour, conversion into liters per hour from tonnage per day will be necessary to help input correct data into speed optimization equations. Where 1 ton is equal to 2831.68 liters. The same process will be utilized in the LFO consumption.

DWT is verified through the Guinea Maritime Profile at the UN Trade & Development hub[5], which is 3,000 DWT. Therefore, these results from this research are theoretical and using operational parameters from this information only illustrates instead of predicting outcomes from North America to Hawaii and the Caribbean. Considering the previous statement, this safely puts the data being used from ship performance clustering dataset in the "less than 10,000 DWT" bracket. Therefore, ship fuel consumption of MGO in that bracket is 240.15 liters per hour and the fuel consumption of LFO is 212.58 liters per hour

Figure 22: Specific Fuel Consumption of ships in the container_performance dataset from ship_fuel_consumption_reporting dataset

	Column1	Diesel /\nGas Oil\n(MDO /\nMGO)	Light Fuel\nOil (LFO)	Number\nof ships	MGO_Consumption_per_Ship(liter/hour)	LFO_Consumption_per_Ship(liter/hour)
0	Containership	4858186.0	10182858.0	5076.0	309.380763	648.468457
1	Less than 10,000 DWT	179784.0	159148.0	242.0	240.146607	212.582055
2	10,000 ? DWT < 15,000	387670.0	573355.0	616.0	203.433403	300.873317
3	15,000 ? DWT < 40,000	848742.0	2347012.0	1634.0	167.905307	464.305726
4	40,000 ? DWT < 80,000	1288696.0	3161549.0	1122.0	371.277307	910.852054
5	80,000 ? DWT < 120,000	1041756.0	1855205.0	668.0	504.115824	897.751678
6	200,000 DWT and above	132894.0	253985.0	139.0	309.051855	590.655225

Source: Author

Bunker_fuel_prices data did not need any changes except to limit the data to the year of 2024. However, the data needs the removal of undesired characters such as "$" and "," in both the MGO and VLSFO columns. These will be replaced with empty spaces. Then the data can be transformed into a float where it will make visualizing a time series of the comparison of fuel prices easier. Once the data was pre-processed then the dataset was checked for duplicates and NAN values, resulting in no duplicates or NAN values.

[5] Data for DWT of Guinea shipping fleets can be found via https://unctadstat.unctad.org/CountryProfile/MaritimeProfile/en-GB/324/index.html

Figure 23: Detection for duplicates and NAN values in bunker_fuel dataset

```
NAN values in the Bunker Fuel dataframe:
 Date                                    0
Month                                   0
Year                                    0
VLSFO Fuel Oil, IMO 2020 Grade, 0.5%    0
Marine Gas Oil                          0
dtype: int64

Number of Duplicates in the Bunker fuel dataframe:  0
```

Source: Author

The important variables to come out of the data will be the minimum speed, the maximum speed, turnaround time in port of a ship, the average operation cost of a ship, the average power consumption to time of a ship, average load percentage of a ship, MGO fuel consumption, and VLSFO consumption.

The minimum and maximum speeds will be values to adhere to when evaluating the speed of a ship inside and outside an ECA. These will be the constraints for a ship to avoid a ship from stalling or from consuming more fuel than needed in an optimized manner, which is in line with Ma et al. (2020) and their model assumptions. Given that the data has established the minimum speeds and maximum speeds, there can now be a range of realistic speeds that the research can work with.

The turnaround time in port is important because it allots time for the ship to dock into port, unload (and reload) containers, and then set sail again. Adding to this point, the ship will constantly be at a capacity of 74.18% when sailing from POL to POD, always making the weight of the ship constant when in motion. This percentage will be at a constant to simulate what the cost would be for a ship that is functioning at only roughly 75% which can be helpful for ship operators to account for their revenue generated by the cargo they carry on their containerships.

The average operational cost is important because this will be the cost to operate a ship from POL to POD, in this case fuel will be separate from this cost. This study will classify fuel as 30% of the total average operational cost because this study will assume a high fixed cost for the ship to run as smoothly as possible, such as quality crew running the ship, insurance for cargo and ship, quality maintenance, etc. Therefore, 70% of the remaining average operational cost will be the fixed costs of the ship. The journeys are similar, but not the same, to the data taken in the container_performance dataset, therefore fuel costs will be variable costs.

The average power consumption to time is important because this will be the power consumption of a containership while it is in transit from POL to POD. Last, but not least, MGO

and VLSFO fuel consumptions are important because these are the values of the fuel consumption for the specific ship that this research is conducting (containership less than 10,000 DWT on long-haul journey).

Figure 24: Important variables of the research

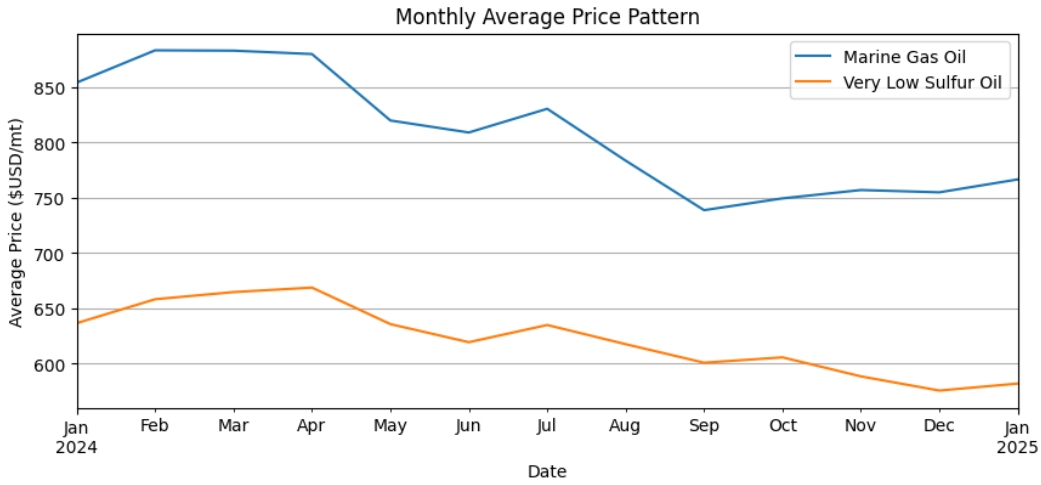
```
The minimum speed a container ship sails is 10.11 knots and the maximum speed a container ship sails is 24.73 knots
The average turnaround time for a container ship is 42.91 hours
The average operation cost, without considering fuel, for a container ship is $187903.64
The average power consumption to time for a container ship is 395878.63 MJ
The MGO consumption per container ship is 240.15 liter/hour
The LFO consumption per container ship is 212.58 liter/hour
The average load percentage for a container ship is 74.18%
```

Source: Author

3.3. Bunker Fuel Prices

Since bunker fuel price is constantly changing, and other studies like those from Fagerholt et al (2015) and Fagerholt and Psaraftis (2015) have only set their fuel price inputs at a constant. This research will continually change the price of bunker fuel on a monthly basis, for the input of the speed optimization equation. This will be done by calculating the time of a route that a ship will take. Starting at the beginning of January 2024, a ship will embark from the POL. If the ship makes it to the POD and the turnaround time doesn't come after the given month's end, the price of fuel will remain the same. However, if the new month begins, the price of fuel will change based on the historical bunker fuel price of that month. Fuel price ratios will also be taken into consideration when determining route choice. As the fuel price ratio between the two fuels gets higher, the higher the cost of fuel will be (Fagerholt and Psaraftis, 2015). This will be important when identifying the manipulated and the controlled variable.

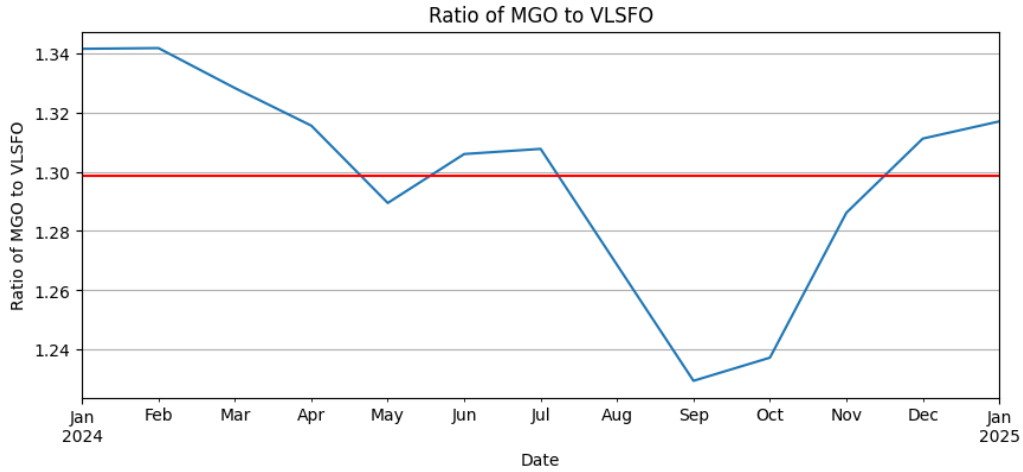
Figure 25: Monthly Average Price Pattern of MGO and VLSFO in 2024



Source: Author

This figure represents the timeseries of the average price of MGO and VLSFO from January 2024 to January 2025. As shown, the graph is experiencing a slight negative trend this year with January average price for MGO and VLSFO at $854.02/mt and $636.61/mt, respectfully, and December average price for MGO and VLSFO at $754.84/mt and $575.68/mt, respectfully. There are many reasons for this trend, scarcity of fuel/oil became less over the year, cost to transport fuel/oil to bunker site decreased over the year, or even the demand for fuel decreased over the year.

Figure 26: Price Ratio between MGO and VLSFO in 2024



Source: Author

This figure represents the timeseries of the price ratio of MGO to VLSFO from January 2024 to January 2025 with the average ratio being 1.298. There is a sizeable dip in the month of September with little increase in October which means that there is a smaller ratio between the two fuels compared to the average. Wherein the beginning of the year, January has a larger ratio compared to the average ratio of fuels. This could be a result of scarcity of fuel/oil becoming less during that month.

3.4. Manipulated and Controlled Journeys

Two scenarios will occur in this study. The first scenario will be the manipulated journey. In this scenario, the ship will choose its journey based on the fuel price ratio of MGO and VLSFO. Based on this fuel price ratio, the trigger for the ship to take the evasion route will be when the ratio is above 1.298, which defines the average ratio between the two fuels in the year 2024. From Fagerholt and Psaraftis (2015), they conducted a sensitivity analysis that describes the increase and decrease of the ratio of cheaper fuel (e.g. HFO) and expensive fuel (e.g. MGO), which demonstrated that the ratio between cheaper fuel and expensive fuel can affect the speed, and therefore the cost, in which a ship travels inside and outside an ECA.

The average ratio will be used rather than the median ratio because both statistics are similar, with median ratio being 1.308, meaning that there is little to no skewness happening in the distribution of the bunker fuel price data. To clarify, the maximum ratio between the two fuels is 1.342 and the minimum ratio is 1.229, making the range between the two ratios 0.113. Therefore, the average represents the best tendency of bunker fuel data. The higher the ratio is to the average, the higher the incentive to take the evasion route because MGO is more expensive than VLSFO and taking the evasion route could limit the costs accrued. If the ratio is below 1.298, then the ship will take the shortest route from POL to POD in the manipulated journey because the ratio will be below the average.

The second scenario is the controlled journey. This will be the shortest route from POL to POD. Regardless of the fuel price ratio, this ship will always take the shortest route. At the end of it all, both journeys will be evaluated on the costs that are accrued.

3.5. Evasion Route

Fuel switching will be a given requirement for this route. From the data, ships have an average load percentage of 74.18% of their maximum capacity, which brings the averages operating cost of the ship, without considering fuel, to $187,903.64. Moreover, ships will be assumed to have the same capacity from POL and POD because ship operators will try to maximize profits when sailing (Fagerholt and Psaraftis, 2015). Unlike in the literature from Fagerholt et al. (2015), Fagerholt and Psaraftis (2015), Langella et al (2016), and Fan et al. (2020), this study will use VLSFO rather than HFO because VLSFO is subject to the most recent

update in regulations and it is compliant with restrictions to SOx emitted outside an ECA (IMOd, 2025).

3.5.1. POL to ECA

The next step is to find the shortest route from the POL to the ECA boundary coordinate points because the ship needs to spend the least amount of time in an ECA as possible and minimize their consumption on expensive MGO. Using the Great Circle Distance formula, the study will take the POL coordinates and calculate the distance between each of the coordinate points listed in the ECA boundary using Python on Google Colab. The coordinate point with the shortest distance from the POL will be the shortest route inside an ECA to limit the amount of expensive MGO fuel consumed.

3.5.2. Speed Optimization

From the POL to the closest ECA boundary coordinate point, there will be an optimized speed given inside the ECA. This is also to limit the consumption of expensive MGO when inside an ECA and will recover time lost by increasing speed and consuming more VLSFO outside an ECA. The speeds will be verified by using the speed_over_ground_knots data in the container_performance dataset. The max speed of a ship in this dataset is 24.73 knots, and the minimum speed is 10.11 knots. The formula that will be used is from Tan et al. (2021) mathematical formula method with adjustments to distance percentage inside and outside of ECAs, as well as fuel consumption of specific fuels:

Table 1: Variable table for the speed optimization equation

Variables for Speed Optimization Equation	Description
$\boldsymbol{V_0^e}$	Speed inside an Emission Control Area
$\boldsymbol{V_0^{ne}}$	Speed outside an Emission Control Area
$\boldsymbol{\overline{v}}$	Maximum speed of a ship
$\boldsymbol{c_0}$	Average operational cost of a ship
$\boldsymbol{D_e}$	Distance percentage of the entire route inside an Emission Control Area
$\boldsymbol{D_{ne}}$	Distance percentage of the entire route outside an Emission Control Area
$\boldsymbol{P_{MGO}}$	Price of Marine Gas Oil
$\boldsymbol{P_{VLSFO}}$	Price of Very Low Sulfur Fuel Oil
$\boldsymbol{F_{MGO}}$	Fuel consumption of Marine Gas Oil
$\boldsymbol{F_{VLSFO}}$	Fuel consumption of Very Low Sulfur Fuel Oil
$\boldsymbol{\alpha}$	Model parameter for a ships operational speed and maximum speed

Source: Tan et al. (2021) and Author

$$V_0^e = \overline{v}\left(\frac{c_0 D_e}{(\alpha - 1)P_{MGO}F_{MGO}}\right)^{\frac{1}{\alpha}}$$

and

$$V_0^{ne} = \overline{v}\left(\frac{c_0 D_{ne}}{(\alpha - 1)P_{VLSFO}F_{VLSFO}}\right)^{\frac{1}{\alpha}}$$

Where V_0^e is the speed (in knots) of a ship inside an ECA; V_0^{ne} is the speed (in knots) of a ship outside an ECA; $\overline{v}$ is the maximum speed (in knots) of a ship; c_0 is the operation cost of a ship ($USD/hour); D_e is the distance percentage of the ship traveling within the ECAs of the entire distance of the route; D_{ne} is the distance percentage the ship travels outside an ECA of the entire distance of the route; P_{MGO} is the price of MGO ($USD/mt); P_{VLSFO} is the price of VLSFO ($USD/mt); F_{MGO} is the fuel consumption of MGO (liter/hour); F_{VLSFO} is the fuel consumption of VLSFO (liter/hour); α is the model parameter which Tan et al. (2021) have taken from Ronen (1982) for the ratio of ship's operational speed and the maximum speed. The authors have said that this parameter is usually 3.

The reason for using this mathematical model is because the variables of the equation can be substituted with data from the datasets that were chosen. Furthermore, with a less complex model, the Python programming language can execute its code faster. Another important note to add clarity is that c_0 is classified as unit operational cost of a ship in Tan et al. (2021), but to get reasonable speeds in this research, the average operational cost, without considering fuel, is used instead. While the equations are not the same to Tan et al. (2021), they are similar and it is important to add D_e/D_{ne} as multiplier to c_0 due to the need to account for the time spent inside and outside the ECA when dealing with average operational cost, therefore giving an accurate speed to sail throughout the ships route.

If the equation were to be kept the same as in the Tan et al. (2021), then the speeds would not reflect reality. For example, using the container_performance dataset, if the POL and the POD were Seattle to Honolulu in January, the speed would be 2.51 knots inside an ECA and 2.89 knots outside an ECA. This is below the minimum speed of a ship in the container_performance dataset, which is below the constraint classified in this research. Therefore, these speeds will not be sufficient. Rather, replacing the unit operation cost (average) with average operation cost multiplied by the distance that a ship sails inside and outside an ECA will reflect real speeds. For example, in the same scenario from Seattle to Honolulu in January, the speed then becomes 10.31 knots inside an ECA and 20.67 knots outside an ECA. This can be verified by Yıldız (2025) stating a minimum speed of 9.97 knots up to 24.5 knots stated by Fan et al. (2020), and of course other speeds in between according to Li et al. (2020), Fagerholt and Psaraftis (2015), Fagerholt et al. (2015), Chen et al. (2018). Therefore, the constraints for speeds given by the container_performance dataset are reasonable speeds.

3.5.3. Evasion of POD ECA

Finally, the evasion strategy will be enacted when the POD is inside another ECA and the ratio of MGO to VLSFO is above the average of 1.289. When this occurs, the POD and ECA coordinates will be tested using Python programming language and the

Great Circle distance formula to find the shortest distance from POD to ECA boundary. The distance from the exit point of the POL ECA and the entry point of the POD ECA will be calculated. Thus, creating the evasion route and completing the evasion route.

3.6. Shortest Route POL to POD

Like the evasion route, fuel switching will be a requirement. This route's priority won't be to exit an ECA via the shortest route from POL to ECA coordinate point, but rather the priority will be to travel the shortest distance from POL to POD. The optimized speed will be calculated the same as the evasion route. The mathematical formula method will come from Tan et al. (2021). Lastly, the evasion of the POD ECA won't be executed because there will be nothing to evade in this scenario.

3.7. Ports and Coordinates

Distances inside the ECA during the shortest route will be verified by BedNBlue sailing distance calculator and by google earth. When there is a discrepancy in distance created by the interactive model, addition or subtraction of distance from model results can correct discrepancy and match reality of actual distance travelled inside the ECA from the POL or POD.

Port coordinates will also be changed to remove any barriers (e.g. land) that would otherwise be in the way of a ship's distance from POL to POD. For example, Port of Seattle coordinates are (47.582418, -122.345709) in reality; however, when calculating the distance from Port of Seattle to Port of Hilo, the distance reflects the most direct path between the two ports without the consideration of land in-between them. Since ships cannot sail over land, the Port of Seattle coordinates were moved to (48.424090, -124.747612). The sailing distance from these coordinates to the actual coordinates for the Port of Seattle was calculated using BedNBlue sailing distance calculator, which were then considered into the equations when calculating the distance from POL to POD. This type of alteration was done to more ports to reflect reality.

3.8. Calculating GFI from Journeys

The biggest piece of data to consider is the data to classify what the emission factors are for MGO and VLSFO fuels. Comer and Osipova (2021) were able to provide detailed figures of types of fuel and emission factors that are correlated with them on a well-to-wake basis for regulatory bodies to determine CO2 emissions. The following table presents their findings:

Table 2: Emission factors for MGO and VLSFO in shipping

Fuel Type	Emissions Factors on a Well-to-Wake basis (g/g)
MGO	3.782
VLSFO	3.734

Source: Comer and Osipova (2021) and Author

Fuel consumption will be accounted for by adding fuel consumed from every route over the year for each fuel type. The sum of respective fuel consumed will then be converted from liters into grams (1L = 1,000g). Then the process to calculate the $GFI_{attained}$ will begin.

Over the course of an annual period and while ships are completing their evasion or shortest routes, summation of costs from operations of ship, costs from fuel purchased, and carbon emissions of ships will be done. The IMO (IMOb, 2025) have developed an equation that calculates the GFI attained from a ship:

Table 3: Variable table for $GFI_{attained}$ equation

Variables for $GFI_{attained}$ Equation	**Description**
$\boldsymbol{GFI_{attained}}$	Greenhouse gas fuel intensity attained over an annual period
$\boldsymbol{EI_j}$	Greenhouse gas intensity of fuel type *j* from a well-to-wake basis
$\boldsymbol{Energy_j}$	Energy consumption of ship using fuel type *j*
$\boldsymbol{Energy_{total}}$	Total energy consumption of a ship
$\boldsymbol{j}$	Type of fuel
$\boldsymbol{J}$	Number of fuels

Source: IMOb (2025) and Author

$$GFI_{attained} = \frac{\Sigma_{j=1}^{J} EI_j \cdot Energy_j}{Energy_{total}}$$

Where $GFI_{attained}$ is the GFI attained over an annual period; j is the type of fuel; J is the number of fuels used over that annual period; EI_j is the GHG intensity (gCO2/MJ) on a well-to-wake basis; $Energy_j$ is the energy consumption (MJ) of ship using j fuel; $Energy_{total}$ is the total energy consumed by the ship such as "electricity delivered from the shore power, and zero-emission energy sources, such as wind propulsion and solar power" (IMOb, 2025).

In this research, it will be assumed that $Energy_j = Energy_{total}$ due to lack of data for total energy consumption of a container ship. For every amount of energy consumed from fuel j, it will be equal to all the energy a ship consumes. This is a significant limitation when considering the true GFI attained because ships also consume energy at port (Aijjou et al., 2019) and renewable energy on ships are fast approaching from the future (Huang et al., 2021). Therefore, the findings of this research will only be hypothetical as the results may not reflect reality. These limitations must be considered when analyzing the results. Further research will be needed on total energy consumption of ships in order to calculate true GFI attained.

The following equation is also from IMOb (2025) which sets the Base and Direct Compliance Target of ship emissions in gCO2/MJ:

Table 4: Variable table for GFI_T equation

Variables for the GFI_T Equation	Description
$\boldsymbol{GFI_T}$	The target for GFI emissions of a ship
$\boldsymbol{Z_T}$	Reduction factor
$\boldsymbol{GFI_{2008}}$	Average GFI in 2008

Source: IMOb (2025) and Author

$$GFI_T = \left(1 - \frac{Z_T}{100}\right) \cdot GFI_{2008}$$

Where GFI_T is the Base target or Direct Compliance target of GFI emissions; Z_T is the reduction factor at an annual rate; and GFI_{2008} is the average GFI of international shipping in 2008 (93.3 gCo2/MJ) as a reference value.

At the end of every annual period, this research will compare their $GFI_{attained}$ to the GFI_T to see if the ship's $GFI_{attained}$ succeeds or fails in complying with two targets: Base Target (BT) and Direct Compliance Target (DCT). Those ships that fail to comply with DCT but succeed in complying with BT are subject to the Tier 1 compliance deficit. Those ships that fail

to comply with BT are subject to Tier 1 and Tier 2 compliance deficits. Tier 1 will be a monetary payment where ships will pay USD $100 per tonne of CO2eq emitted on the basis of well-to-wake. Tier 2 will be the choice of one or a combination of three options: diversion of surplus units acquired by other ships, surplus units from a previous reporting period, or a monetary payment of USD $380 per tonne of CO2eq emitted on the basis of well-to-wake. Those ships that do succeed in complying with DCT will be subject to compensation via surplus units from the IMO Net-Zero Fund (IMOb, 2025). In this study, it will be assumed that failure to meet BT will result in monetary payment.

The following equations are from IMOb (2025) and represent the deficits for ships that fail to meet DCT but succeed to meet BT and ships that fail to meet BT:

Table 5: Variable table for the tier deficit compliance equation

Variable for the Tier Deficit Compliance Equations	Description
$\boldsymbol{GFI_{BT}}$	Base target greenhouse gas fuel intensity
$\boldsymbol{GFI_{DCT}}$	Direct compliance target greenhouse gas fuel intensity

Source: Author

If

$$GFI_{DCT} < GFI_{attained} \leq GFI_{BT}$$

$$\text{Tier 1 Deficit Compliance} = (GFI_{DCT} - GFI_{attained}) \cdot Energy_{total}$$

Or if

$$GFI_{attained} > GFI_{BT}$$

$$\text{Tier 1 Deficit Compliance} = (GFI_{DCT} - GFI_{BT}) \cdot Energy_{total}$$

and

$$\text{Tier 2 Deficit Compliance} = (GFI_{BT} - GFI_{attained}) \cdot Energy_{total}$$

Where GFI_{DCT} is the direct compliance target for GFI on a well-to-wake basis in an annual reporting period; GFI_{BT} is the base target for GFI on a well-to-wake basis in an annual reporting period.

3.9. Evaluation of Journeys

This is the part of the research that takes account of the findings made by the manipulated journey and the controlled journey. The manipulated and controlled journeys have accrued their costs on an annual basis and have calculated the GFI that was attained over that year. Each journey will be compared to each other to see if taking the manipulated journey is more expensive than taking the controlled journey, or vice versa, based on new IMO regulations.

In terms of how cost will be calculated, fuel costs, operating costs without considering fuel, and cost from over-production of GFI attained will be used to compare between the manipulated and controlled journeys. Fuel costs will change over the course of the year. Operating costs will remain constant over the year because the average load capacity will be assumed to be the same over the year. Finally, costs from GFI attained will be incorporated at the end of the year when calculations for GFI attained are compared to BT and DCTs.

4. Results and Analysis

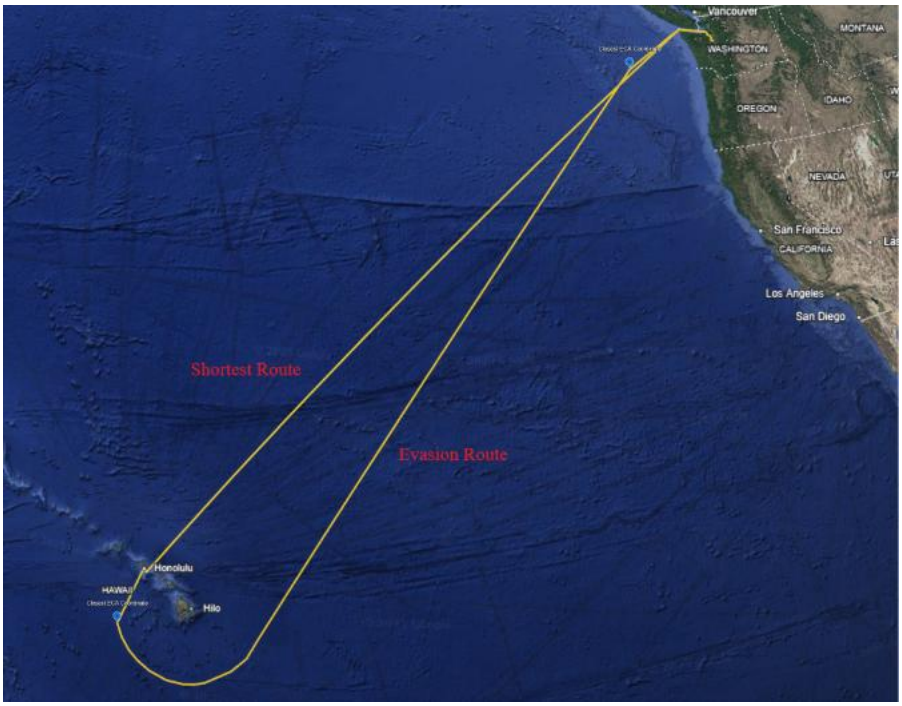
This section is for the results that were found using the interactive model. The first step is to define the hypothetical routes that were sailed by ships to demonstrate the distances from POL to POD and the percentage of distance inside an ECA and outside an ECA. To help with the challenges of the model be more accurate, google earth was used to verify or adjust any distances for the shortest routes in the model. The second step is to implement the distance percentages into the speed formula and simulate the number of times a ship completes its route in 2024 while the price of fuels changes every month and turnaround time at port is considered. Finally, the energy consumed and fuel consumed on each route was totaled and put into CO2 emissions, and the GFI attained equations to calculate the remedial units owed to the Net-Zero fund of the IMO. The GFI target will be GFI 2028 (BT = 89.57 gCO2/MJ) as it is the year when the regulation on ships will commence. However, this research can substitute any target year with this interactive model.

For the null hypothesis of this research, the average cost of the controlled journeys will have no significant difference between the average cost of manipulated journeys ($\mu_1 = \mu_2$). For the alternative hypothesis, the average cost of the controlled journeys will have a significant cost difference between the average cost of the manipulated journeys ($\mu_1 \neq \mu_2$). In this instance, a two-sample t-test will be conducted to determine statistical significance between the two journeys. This test will determine if the difference in average costs is substantial or not.

4.1. Seattle to Honolulu

The following figure and tables are the results that were produced from the interactive model of a ships journey from Seattle to Honolulu and the visualization of the routes on Google Earth:

Figure 27: Evasion Route and Shortest Route from Seattle to Honolulu



Source: Author

Table 6, 7, 8: Total cost comparison of two types of Journeys from Seattle to Honolulu

Route	Total Cost (USD)
Seattle to Honolulu (Manipulated Journey)	9,488,168.89
Seattle to Honolulu (Controlled Journey)	9,464,611.66

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Seattle to Honolulu (Manipulated Journey)	36	303.98	1715.35	4291.96	15439266.46	389.09
Seattle to Honolulu (Controlled Journey)	39	302.28	1583.06	4375.11	15439266.46	385.91

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Seattle to Honolulu (Manipulated Journey)	18,726.29	1,757,287.45	7,328,241.96	130,379.24	253,533.96
Seattle to Honolulu (Controlled Journey)	18,726.29	1,738,614.99	7,328,241.96	120,190.91	258,837.51

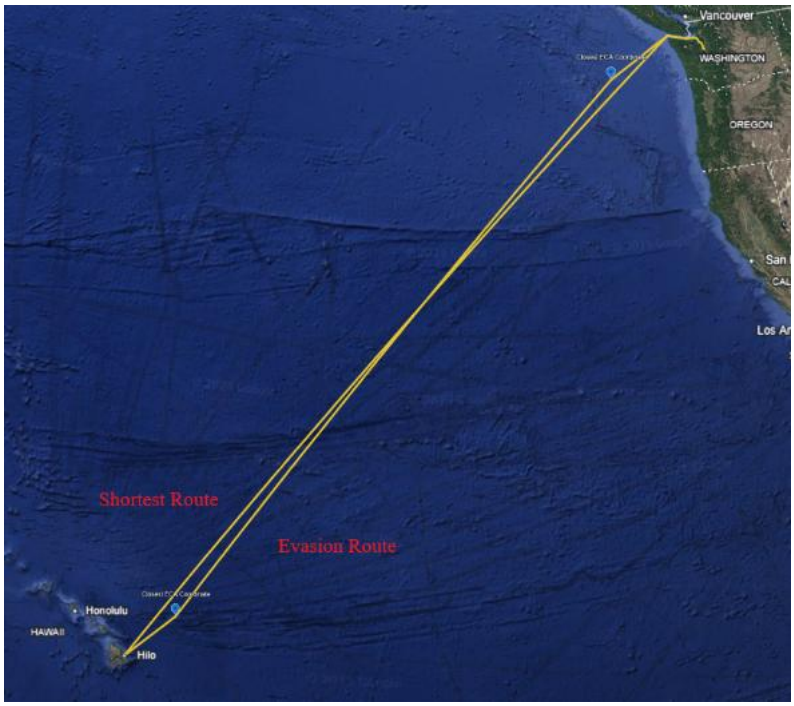
Source: Author

The results of this route and journey types show that the controlled journey will cost a ship operator less than the manipulated journey when only considering the operating cost of a ship, fuel costs, and the remedial unit costs when the ship's GFI attained is more than the GFI target. The difference between the two journeys is $23,557.23. The GFI attained for the manipulated journey is more than the GFI attained from the controlled journey with a difference of 3.18 gCO2/MJ. Both GFI's attained are over the GFI target, however if a ship were to take the controlled journey from Seattle to Honolulu, then the GFI attained would be less than the manipulated journey. The total number of routes completed in the controlled journey is more than the total number of routes completed in the manipulated journey with a difference of 3 more journeys completed.

4.2. Seattle to Hilo

The following figure and tables are the results that were produced from the interactive model of a ships journey from Seattle to Hilo and the visualization of the routes on Google Earth:

Figure 28: Evasion Route and Shortest Route from Seattle to Hilo



Source: Author

Table 9, 10, 11: Total cost comparison of two types of Journeys from Seattle to Honolulu

Route	Total Cost (USD)
Seattle to Hilo (Manipulated Journey)	11,057,687.79
Seattle to Hilo (Controlled Journey)	11,279,498.64

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Seattle to Hilo (Manipulated Journey)	44	279.91	2015.04	3571.44	19398052.73	287.99
Seattle to Hilo (Controlled Journey)	50	285.32	2046.57	3646.80	19793931.36	287.63

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Seattle to Hilo (Manipulated Journey)	18,726.29	1,757,287.45	7,328,241.96	130,379.24	253,533.96
Seattle to Hilo (Controlled Journey)	24,008.06	1,489,778.02	9,395,182.00	155,052.78	215,477.78

Source: Author

Unlike the analysis of Seattle to Honolulu, the manipulated journey from Seattle to Hilo will cost a ship operator less than the controlled journey with a difference of $221,810.85. However, the GFI attained by the controlled journey was slightly less than the manipulated journey by 0.36 gCO2/MJ. Both journeys still have a high GFI attained compared to the GFI target for 2028 and the difference between both GFI attained is small. In any case, the controlled journey completed more routes completed annually than the manipulated journey with a difference of 6 more completed routes. In any case, given that the cost difference is significantly high between the journeys, GFI attained is relatively the same, and the difference of total number of routes completed between journeys is 6.

4.3. Seattle to Kahului

The following figure and tables are the results that were produced from the interactive model of a ships journey from Seattle to Kahului and the visualization of the routes on Google Earth:

Figure 29: Evasion Route and Shortest Route from Seattle to Kahului

Source: Author

Table 12, 13, 14: Total cost comparison of two types of Journeys from Seattle to Kahului

Route	Total Cost (USD)
Seattle to Kahului (Manipulated Journey)	11,265,210.82
Seattle to Kahului (Controlled Journey)	11,264,389.74

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Seattle to Kahului (Manipulated Journey)	45	283.49	2059.71	3600.61	19793931.36	285.96
Seattle to Kahului (Controlled Journey)	50	285.32	2056.19	3602.41	19793931.36	287.63

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Seattle to Kahului (Manipulated Journey)	24,008.06	1,477,216.97	9,395,182.00	156,056.87	212,746.93
Seattle to Kahului (Controlled Journey)	24,008.06	1,476,563.56	9,395,182.00	155,781.10	212,855.03

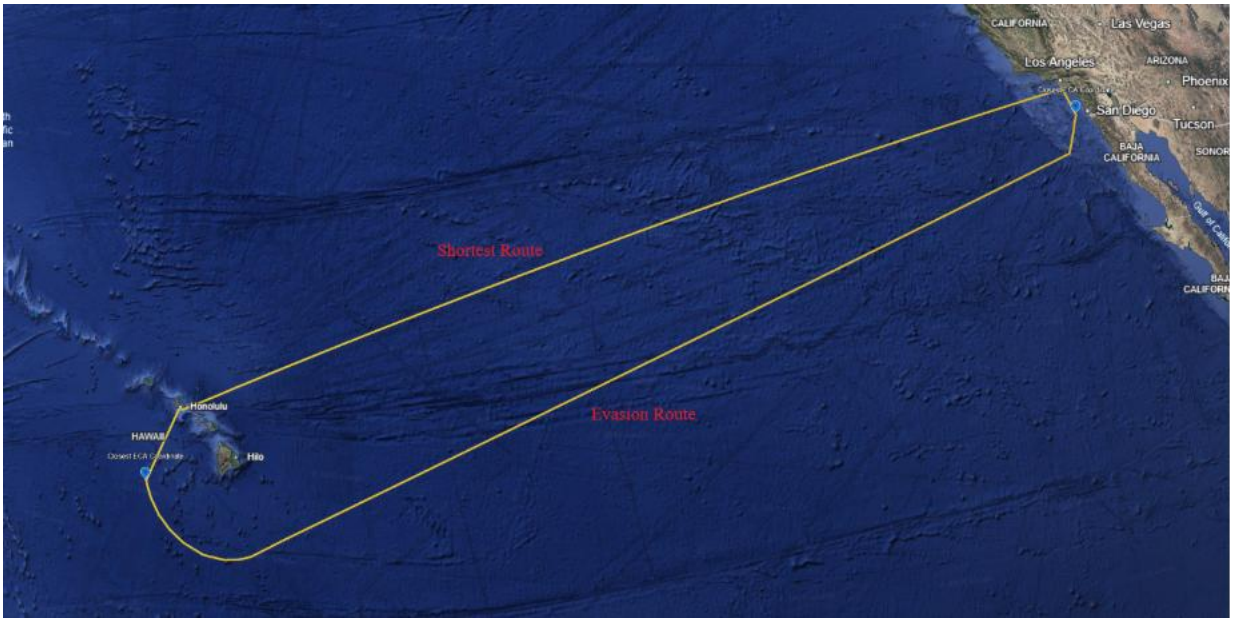
Source: Author

The total cost difference from both journeys is very close, but the controlled journey from Seattle to Kahului will cost a ship operator less than the manipulated journey with a difference of $821.08. Most noticeably though, the GFI attained from the manipulated journey is less than the GFI attained from controlled journey, with a difference of 1.67 gCO2/MJ, which could be a result from the total number of routes completed being less than the total number of routes completed in the controlled journey. The difference between completed routes is 5 completed routes.

4.4. Los Angeles to Honolulu

The following figure and tables are the results that were produced from the interactive model of a ships journey from Los Angeles to Honolulu and the visualization of the routes on Google Earth:

Figure 30: Evasion Route and Shortest Route from Los Angeles to Honolulu



Source: Author

Table 15, 16, 17: Total cost comparison of two types of Journeys from Los Angeles to Honolulu

Route	Total Cost (USD)
Los Angeles to Honolulu (Manipulated Journey)	10,087,508.17
Los Angeles to Honolulu (Controlled Journey)	9,891,064.48

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Los Angeles to Honolulu (Manipulated Journey)	39	298.41	972.53	4834.91	17022780.97	341.16
Los Angeles to Honolulu (Controlled Journey)	42	296.90	847.65	4915.38	16626902.34	346.61

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Los Angeles to Honolulu (Manipulated Journey)	20,646.93	1,627,442.27	8,079,856.52	73,968.02	285,594.42
Los Angeles to Honolulu (Controlled Journey)	20,166.77	1,624,040.12	7,891,952.88	64,275.07	290,629.63

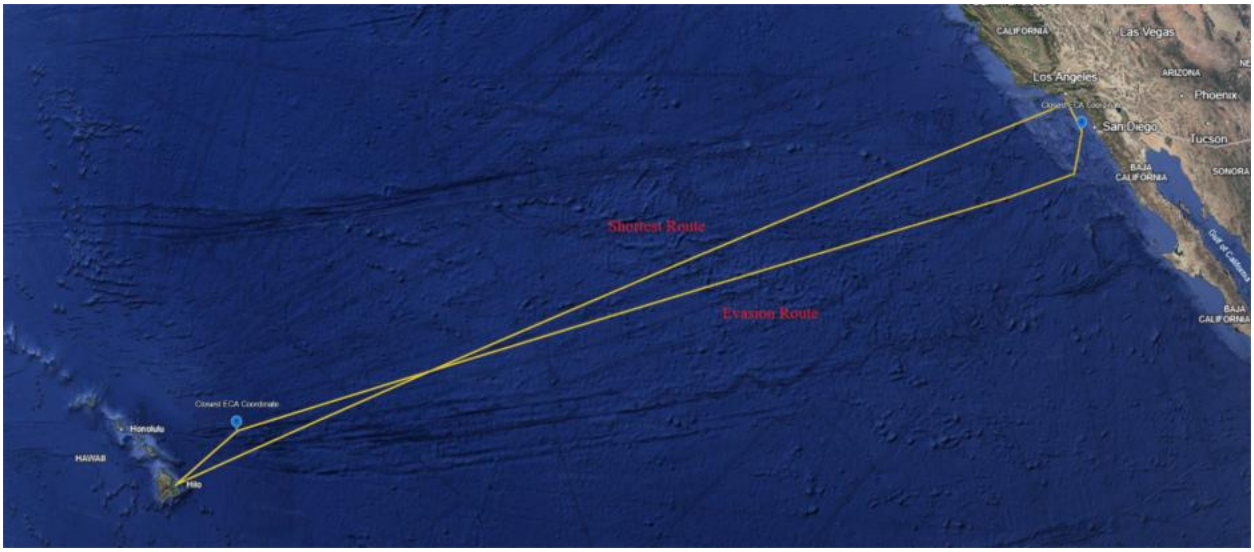
Source: Author

The controlled journey from Los Angeles to Honolulu will cost a ship operator less than the manipulated journey in the given year with a difference of $196,443.69. Both GFI's attained are substantially high with the manipulated journey contributing less GFI attained than the controlled journey with a difference of 5.45 gCO2/MJ. Nevertheless, the controlled journey completed the most routes in a year with a difference of 3 completed routes.

4.5. Los Angeles to Hilo

The following figure and tables are the results that were produced from the interactive model of a ships journey from Los Angeles to Hilo and the visualization of the routes on Google Earth:

Figure 31: Evasion Route and Shortest Route from Los Angeles to Hilo



Source: Author

Table 18, 19, 20: Total cost comparison of two types of Journeys from Los Angeles to Hilo

Route	Total Cost (USD)
Los Angeles to Hilo (Manipulated Journey)	11,346,305.05
Los Angeles to Hilo (Controlled Journey)	11,352,955.37

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Los Angeles to Hilo (Manipulated Journey)	47	279.88	1167.92	4311.14	20189809.99	271.38
Los Angeles to Hilo (Controlled Journey)	51	281.64	1062.04	4437.20	20189809.99	272.38

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Los Angeles to Hilo (Manipulated Journey)	24,488.22	1,394,865.87	9,583,085.64	88,909.96	254,955.36
Los Angeles to Hilo (Controlled Journey)	24,488.22	1,402,534.48	9,583,085.64	80,536.06	262,310.98

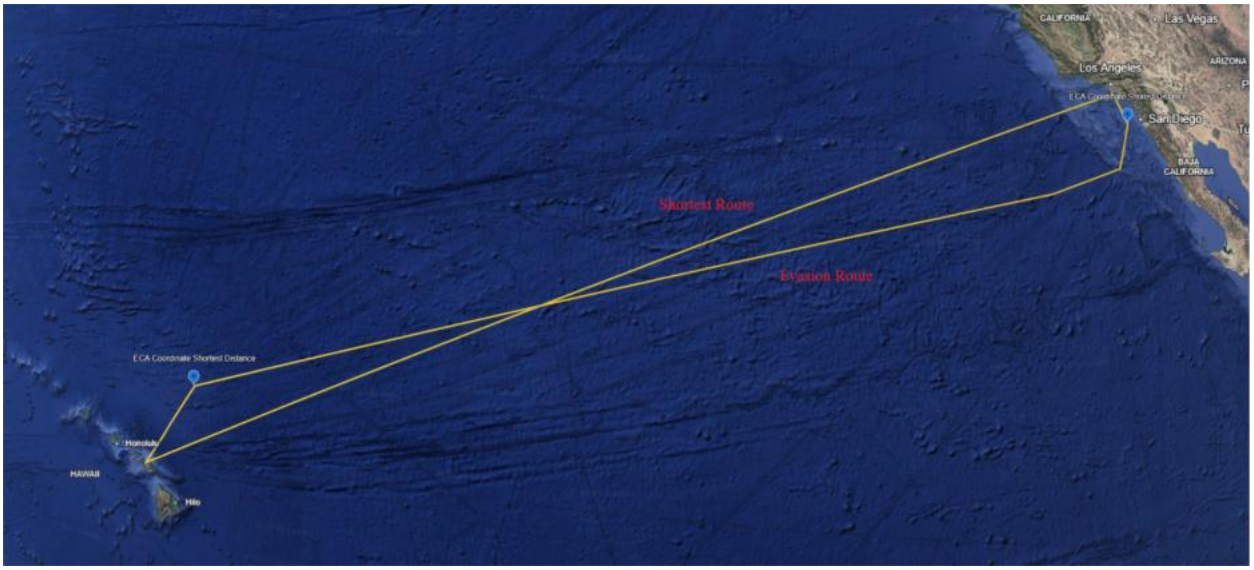
Source: Author

For Los Angeles to Hilo, the manipulated journey will cost less for a ship operator than the controlled journey. However, there is not much difference between the total cost with only a difference of $6,650.32. The manipulated journey has less GFI attained than the controlled journey but only a difference of 1.0 gCO2/MJ. The controlled journey, once again, has more completion of routes throughout a given year, with a difference of 4 completed journeys.

4.6. Los Angeles to Kahului

The following figure and tables are the results that were produced from the interactive model of a ships journey from Los Angeles to Kahului and the visualization of the routes on Google Earth:

Figure 32: Evasion Route and Shortest Route from Los Angeles to Kahului



Source: Author

Table 21, 22, 23: Total cost comparison of two types of Journeys from Los Angeles to Kahului

Route	Total Cost (USD)
Los Angeles to Kahului (Manipulated Journey)	11,179,285.73
Los Angeles to Kahului (Controlled Journey)	11,188,642.86

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Los Angeles to Kahului (Manipulated Journey)	46	280.96	1151.00	4346.52	19793931.36	277.74
Los Angeles to Kahului (Controlled Journey)	50	283.02	1044.29	4479.12	19793931.36	279.05

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Los Angeles to Kahului (Manipulated Journey)	24,008.06	1,415,354.23	9,395,182.00	87,584.91	257,156.52
Los Angeles to Kahului (Controlled Journey)	24,008.06	1,425,192.05	9,395,182.00	79,248.42	265,012.32

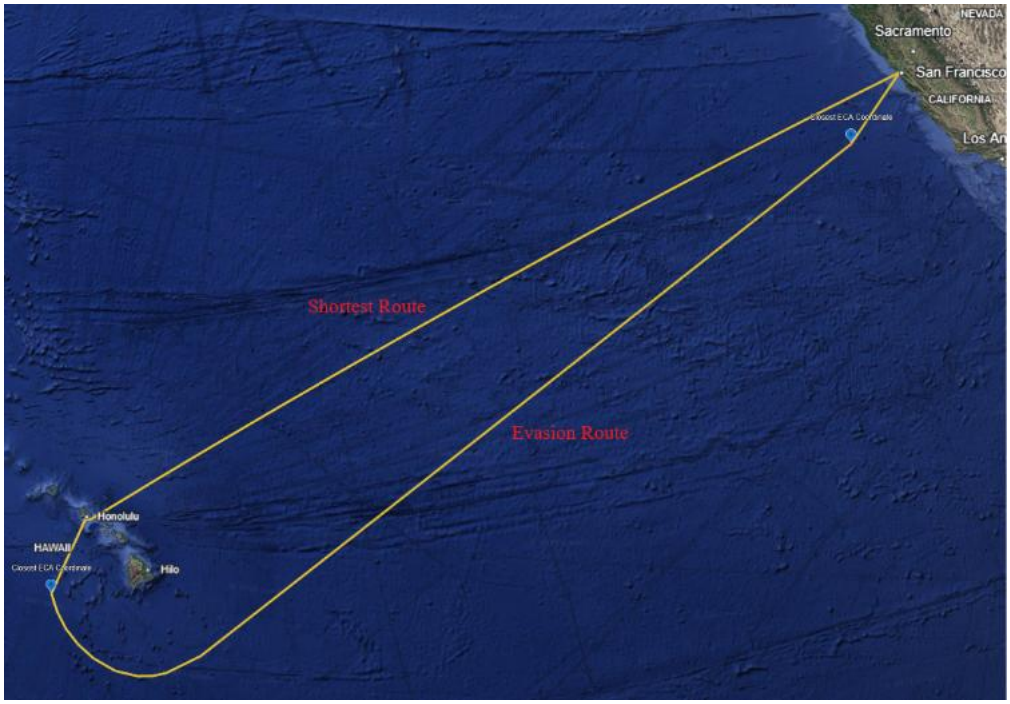
Source: Author

The manipulated journey of Los Angeles to Kahului will cost a ship operator less than the controlled journey by a difference of $9,357.13. The controlled journey has a higher GFI attained than the manipulated journey with a difference of 1.31 gCO2/MJ, making the controlled journey slightly more environmentally friendly. This can be helpful as well because the completion of routes for the controlled journey is greater than the completion of routes for the manipulated journey with a difference of 4 completed routes.

4.7. Oakland to Honolulu

The following figure and tables are the results that were produced from the interactive model of a ships journey from Oakland to Honolulu and the visualization of the routes on Google Earth:

Figure 33: Evasion Route and Shortest Route from Oakland to Honolulu



Source: Author

Table 24, 25, 26: Total cost comparison of two types of Journeys from Oakland to Honolulu

Route	Total Cost (USD)
Oakland to Honolulu (Manipulated Journey)	10,282,723.00
Oakland to Honolulu (Controlled Journey)	10,262,387.49

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Oakland to Honolulu (Manipulated Journey)	40	295.68	1573.49	4257.70	17418659.60	334.77
Oakland to Honolulu (Controlled Journey)	44	294.35	1442.06	4347.21	17418659.60	332.36

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Oakland to Honolulu (Manipulated Journey)	21,127.09	1,622,996.34	8,267,760.16	119,461.09	251,378.32
Oakland to Honolulu (Controlled Journey)	21,127.09	1,607,063.16	8,267,760.16	109,364.84	257,072.24

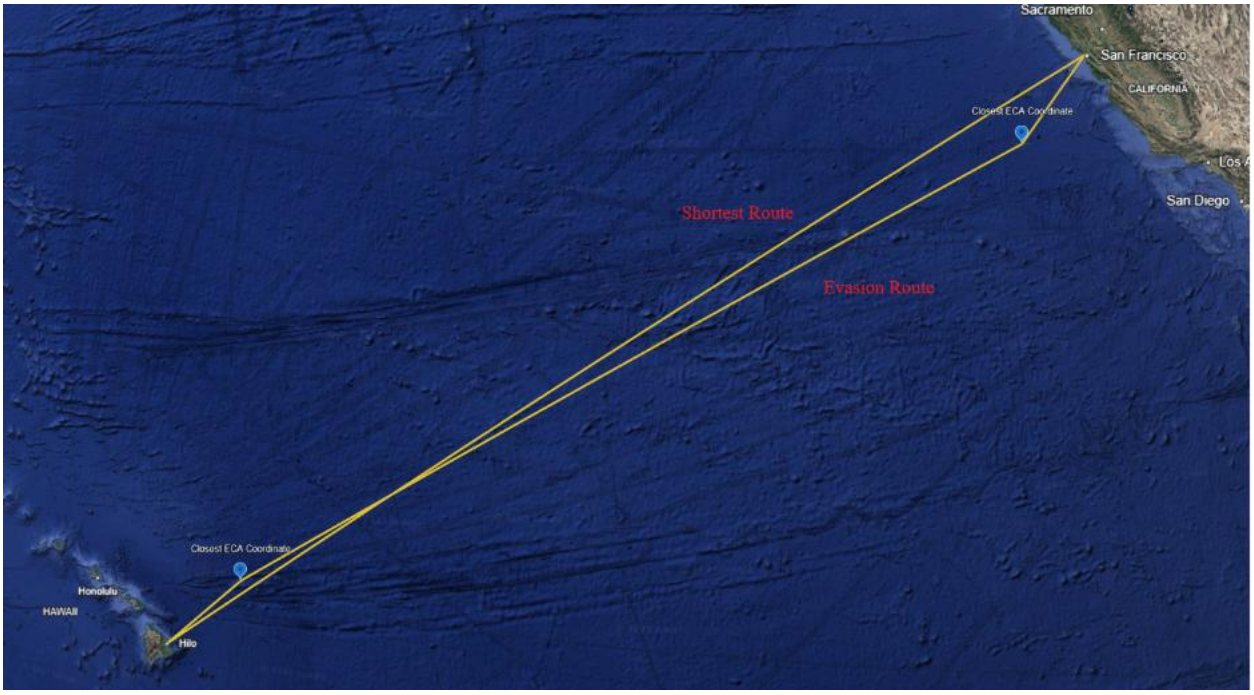
Source: Author

From Oakland to Honolulu, the controlled journey will cost the ship operator less than the manipulated journey with a difference of $20,335.51. GFI attained is extremely high in both journeys; however, the controlled journey has less GFI attained than the manipulated journey with a difference of 2.41 gCO2/MJ. The number of routes completed in the controlled journey is higher than the number of routes completed in the manipulated journey with a difference of 4 completed routes.

4.8. Oakland to Hilo

The following figure and tables are the results that were produced from the interactive model of a ships journey from Oakland to Hilo and the visualization of the routes on Google Earth:

Figure 34: Evasion Route and Shortest Route from Oakland to Hilo



Source: Author

Table 27, 28, 29: Total cost comparison of two types of Journeys from Oakland to Hilo

Route	Total Cost (USD)
Oakland to Hilo (Manipulated Journey)	12,201,066.39
Oakland to Hilo (Controlled Journey)	12,199,870.59

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Oakland to Hilo (Manipulated Journey)	51	271.35	1893.06	3514.85	22169203.12	243.94
Oakland to Hilo (Controlled Journey)	56	271.25	1886.72	3518.56	22169203.12	243.82

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Oakland to Hilo (Manipulated Journey)	26,889.03	1,300,457.38	10,522,603.84	143,419.08	207,697.06
Oakland to Hilo (Controlled Journey)	26,889.03	1,299,459.65	10,522,603.84	142,899.81	208,018.26

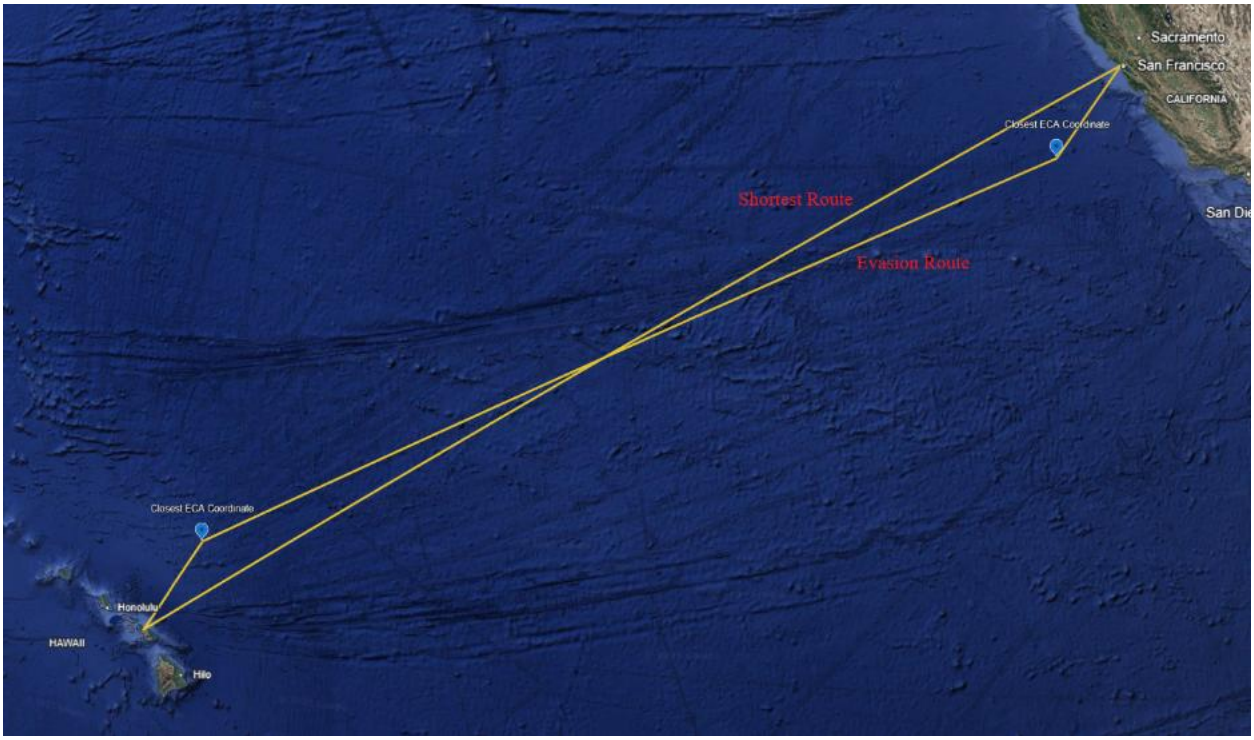
Source: Author

The controlled journey from Oakland to Hilo will cost a ship operator less than the manipulated journey with a difference of $1,195.80. The GFI attained from both journeys is relatively even with a slight edge that favors the controlled journey with a difference of 0.12 gCO2/MJ. Not to mention that the controlled journey has completed more routes than the manipulated journey.

4.9. Oakland to Kahului

The following figure and tables are the results that were produced from the interactive model of a ships journey from Oakland to Kahului and the visualization of the routes on Google Earth:

Figure 35: Evasion Route and Shortest Route from Oakland to Kahului



Source: Author

Table 30, 31, 32: Total cost comparison of two types of Journeys from Oakland to Kahului

Route	Total Cost (USD)
Oakland to Kahului (Manipulated Journey)	12,027,596.70
Oakland to Kahului (Controlled Journey)	12,023,521.89

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Oakland to Kahului (Manipulated Journey)	50	271.62	1872.73	3537.79	21773324.49	248.49
Oakland to Kahului (Controlled Journey)	55	271.29	1855.32	3546.67	21773324.49	248.10

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Oakland to Kahului (Manipulated Journey)	26,408.87	1,314,922.70	10,334,700.20	142,178.17	209,386.76
Oakland to Kahului (Controlled Journey)	26,408.87	1,311,681.38	10,334,700.20	140,810.04	209,921.40

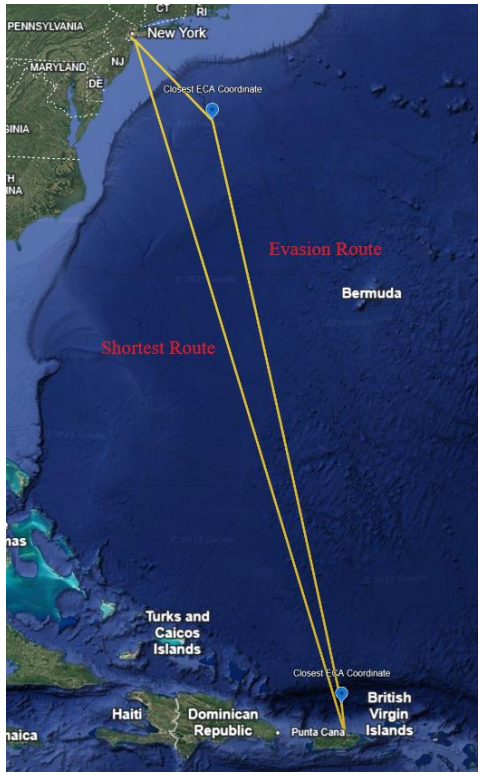
Source: Author

The controlled journey from Oakland to Kahului will cost a ship operator less than the manipulated journey with a difference of $4,074.81. The GFI attained from the controlled journey is less than the GFI attained from the manipulated journey with a difference of 0.39 gCO2/MJ. Furthermore, the controlled journey completed more routes than the manipulated journey.

4.10. New York to San Juan

The following figure and tables are the results that were produced from the interactive model of a ships journey from New York to San Juan, Puerto Rico and the visualization of the routes on Google Earth:

Figure 36: Evasion Route and Shortest Route from New York to San Juan



Source: Author

Table 33, 34, 35: Total cost comparison of two types of Journeys from New York to San Juan

Route	Total Cost (USD)
New York to San Juan (Manipulated Journey)	14,738,107.93
New York to San Juan (Controlled Journey)	14,719,562.32

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
New York to San Juan (Manipulated Journey)	66	242.61	1589.41	3232.87	28503261.16	169.18
New York to San Juan (Controlled Journey)	72	241.49	1465.89	3319.42	28503261.16	167.89

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
New York to San Juan (Manipulated Journey)	34,571.61	862,331.50	13,529,062.08	120,933.52	191,209.22
New York to San Juan (Controlled Journey)	34,571.61	848,284.86	13,529,062.08	111,221.26	196,422.52

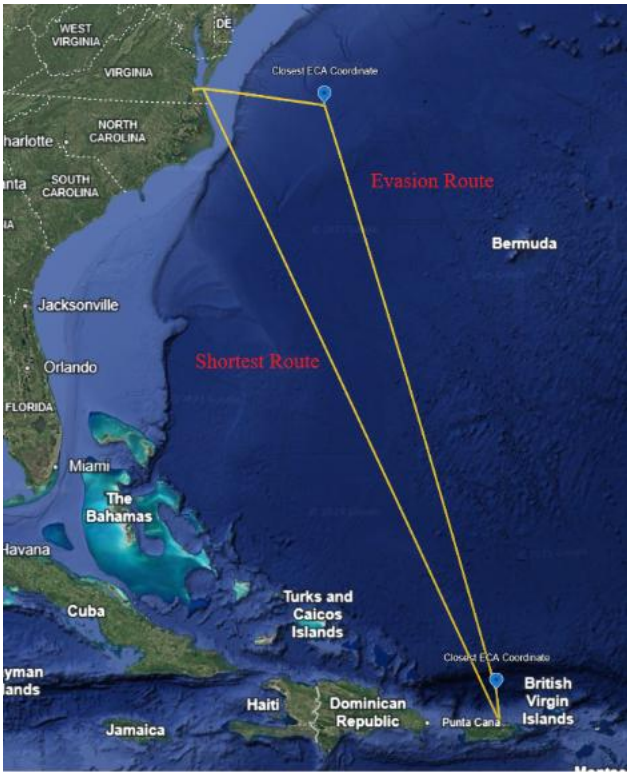
Source: Author

From New York to San Juan, the controlled journey will cost less than the manipulated journey by a difference of $18,545.61. The GFI attained from the controlled journey is also less with a difference of 1.29 gCO2/MJ. The total number of completed journeys for the controlled route is also higher than the evasion route with a difference of 6 completed routes.

4.11. Virginia to San Juan

The following figure and tables are the results that were produced from the interactive model of a ships journey from Virginia to San Juan, Puerto Rico and the visualization of the routes on Google Earth:

Figure 37: Evasion Route and Shortest Route from Virginia to San Juan



Source: Author

Table 36, 37, 38: Total cost comparison of two types of Journeys from Virginia to San Juan

Route	Total Cost (USD)
Virginia to San Juan (Manipulated Journey)	15,080,931.63
Virginia to San Juan (Controlled Journey)	15,063,274.84

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Virginia to San Juan (Manipulated Journey)	68	241.20	1660.38	3143.86	29295018.41	164.00
Virginia to San Juan (Controlled Journey)	74	240.12	1543.66	3225.33	29295018.41	162.79

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Virginia to San Juan (Manipulated Journey)	35,531.93	828,532.70	13,904,869.36	126,201.07	185,796.57
Virginia to San Juan (Controlled Journey)	35,531.93	815,134.10	13,904,869.36	117,046.77	190,692.67

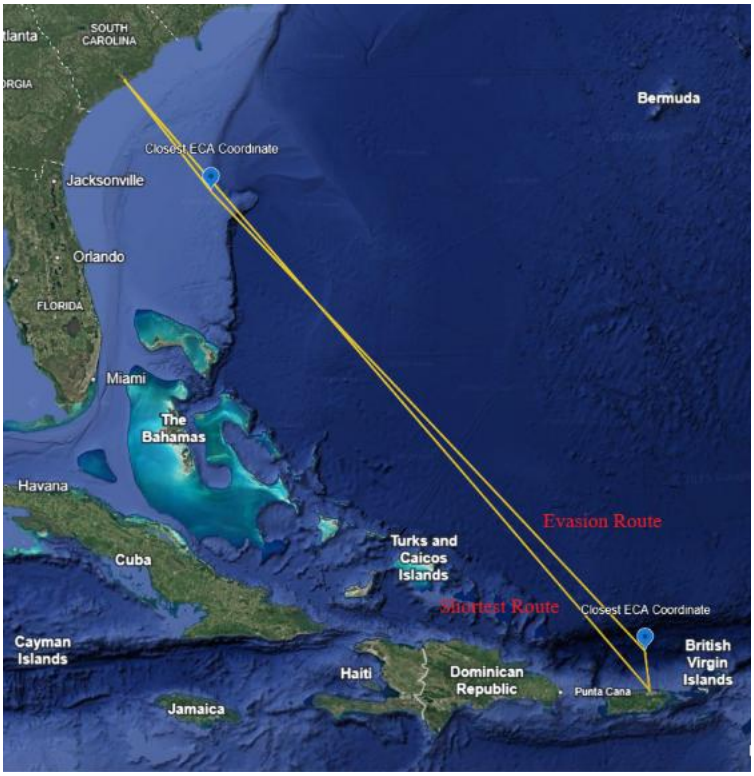
Source: Author

Once again, the controlled journey from Virginia to San Juan will cost a ship operator less than the manipulated journey with a difference of $17,656.79. The controlled journey also has produced less GFI attained over the year by a difference of 1.21 gCO2/MJ. Like previous routes from New York to San Juan, the total number of completed routes of the controlled journey is significantly higher than the routes completed in the manipulated journey.

4.12. Charleston to San Juan

The following figure and tables are the results that were produced from the interactive model of a ships journey from Charleston to San Juan, Puerto Rico and the visualization of the routes on Google Earth:

Figure 38: Evasion Route and Shortest Route from Charleston to San Juan



Source: Author

Table 39, 40, 41: Total cost comparison of two types of Journeys from Charleston to San Juan

Route	Total Cost (USD)
Charleston to San Juan (Manipulated Journey)	16,354,518.88
Charleston to San Juan (Controlled Journey)	16,349,823.73

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Charleston to San Juan (Manipulated Journey)	75	226.23	1654.49	2863.79	32462047.43	139.19
Charleston to San Juan (Controlled Journey)	82	225.86	1633.41	2875.14	32462047.43	138.89

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Charleston to San Juan (Manipulated Journey)	39,373.22	612,074.02	15,408,098.48	125,527.92	169,445.24
Charleston to San Juan (Controlled Journey)	39,373.22	608,375.02	15,408,098.48	123,870.12	170,106.89

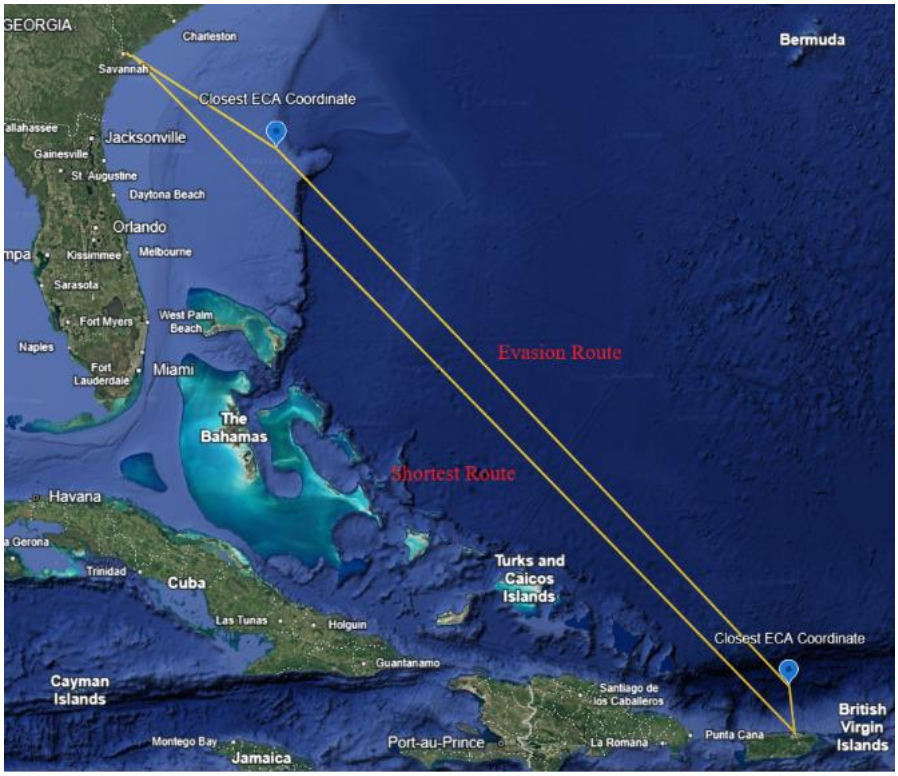
Source: Author

The controlled journey will cost a ship operator less than the manipulated journey from Charleston to San Juan by a difference of $4,695.15. GFI attained from the controlled journey is lower, as well, by a small difference of 0.3 gCO2/MJ. Nevertheless, the total number of routes completed from the controlled journey is more than the total number of routes completed from the manipulated journey, with a difference of 7 more routes completed in the controlled journey.

4.13. Savannah to San Juan

The following figure and tables are the results that were produced from the interactive model of a ships journey from Savannah to San Juan, Puerto Rico and the visualization of the routes on Google Earth:

Figure 39: Evasion Route and Shortest Route from Savannah to San Juan



Source: Author

Table 42, 43, 44: Total cost comparison of two types of Journeys from Savannah to San Juan

Route	Total Cost (USD)
Savannah to San Juan (Manipulated Journey)	16,004,228.39
Savannah to San Juan (Controlled Journey)	16,190,561.80

Route	Total number of Routes completed (Annually)	Days at Sea (Annually)	Annual MGO CO2 emissions (tonnes)	Annual VLSFO CO2 emissions (tonnes)	Annual Total Energy (MJ)	Annual GFI Attained (gCO2/MJ)
Savannah to San Juan (Manipulated Journey)	73	226.02	1669.54	2846.70	31670290.17	142.60
Savannah to San Juan (Controlled Journey)	81	227.96	1603.48	2941.47	32066168.80	141.74

Route	Remedial Units - Tier 1 (USD)	Remedial Units - Tier 2 (USD)	Annual Operating Cost (USD)	Annual Cost MGO (USD)	Annual Cost VLSFO (USD)
Savannah to San Juan (Manipulated Journey)	38,412.89	638,243.68	15,032,291.20	126,878.17	168,402.44
Savannah to San Juan (Controlled Journey)	38,893.06	635,680.92	15,220,194.84	121,710.81	174,082.18

Source: Author

The manipulated journey for a ship operator will cost less than the controlled journey from Savannah to San Juan with an extreme difference of $186,333.41. However, GFI attained is smaller from the controlled journey and the number of routes completed from the controlled journey is more than the manipulated journey, with a difference of 0.86 gCO2/MJ and 8 routes completed, respectively.

4.14. Results and Analysis Conclusion

Table 45: Cost-effective journey outcome

Type of Journey	Number of Cost-effective Journey Outcomes
Manipulated Journey	4
Controlled Journey	9

Source: Author

Given that the number of cost-effective journey outcomes favor the controlled journey more (this is determined by the cost difference between both journeys in each hypothetical route), there is still the question of significance when comparing the two journeys on a statistical level. When comparing the two journeys, the two-sample t-test was utilized and showed that there was no significant difference between the average cost of the two journeys in both the Pacific and Atlantic.

Figure 40 and 41: Two-sample t-test of the average cost difference between the controlled journey and the manipulated journey in the Pacific and Atlantic

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--- Hypothesis Test Results (Pacific) ---

Two-sample t-test: t = 0.002, α = 0.0017, p = 0.9983 → Fail to reject null: no significant cost difference.

--- Hypothesis Test Results (Atlantic) ---

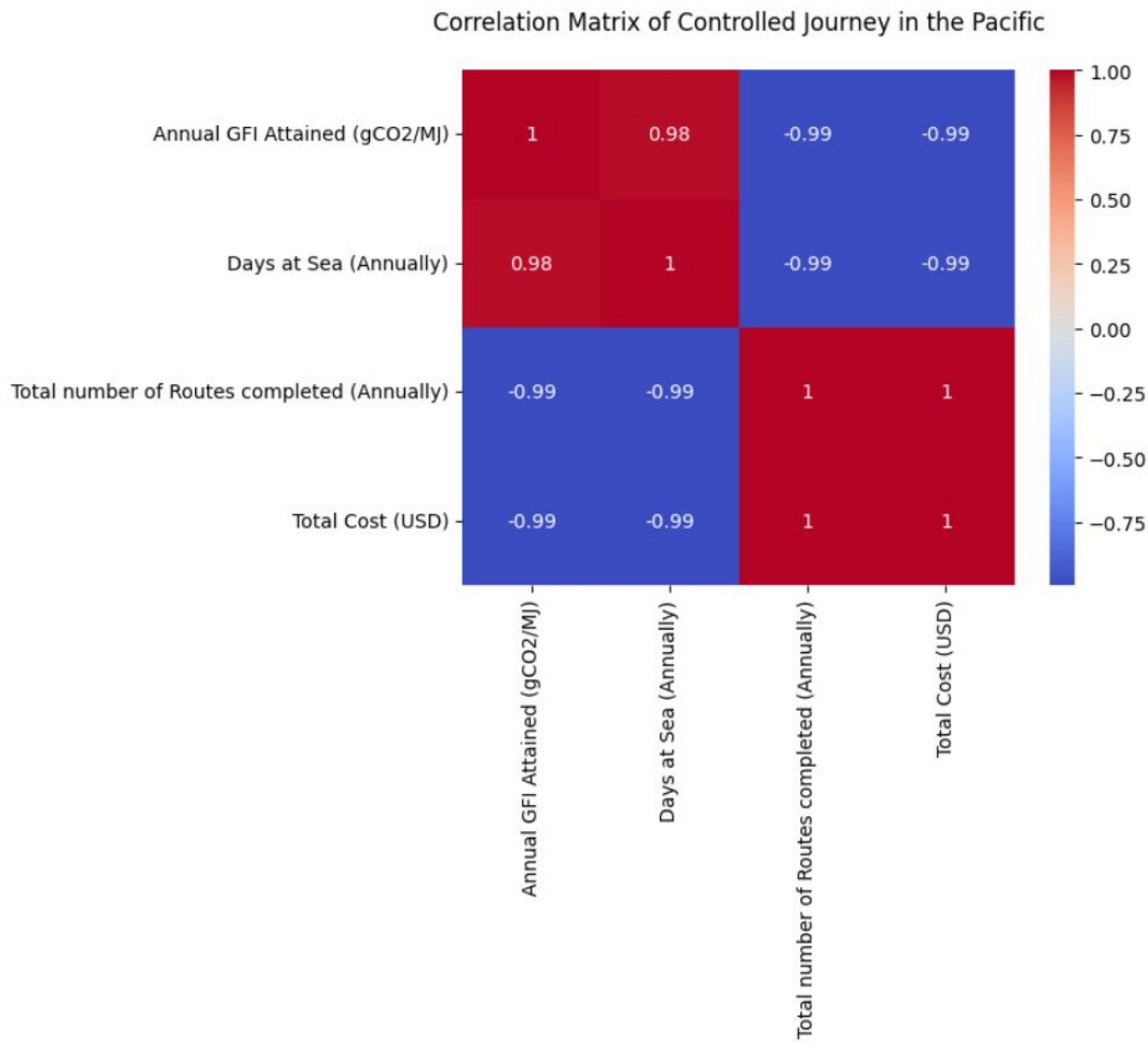
Two-sample t-test: t = -0.065, α = 0.0500, p = 0.9500 → Fail to reject null: no significant cost difference.
```

Source: Author

This might be a result of a small sample size as there are only 9 scenarios in the pacific and 4 scenarios in the Atlantic. An increase in the number of scenarios from POLs to PODs in each region might change the results of the significance test. Nevertheless, there is no significant difference in cost between the two journeys in the Pacific and the Atlantic in this research. The following figure shows that there is a very strong inverse correlation between annual GFI attained and the number of routes completed. This means that the more routes completed in a year, the lower the GFI attained on an annual basis. Therefore, the shorter routes or less time

spent sailing will be beneficial for ship operators. This also supports Table 45 for the best journey outcome in terms of cost.

Figure 42: Correlation Matrix of the Controlled Journey in the Pacific



This correlation is the same (or similar) for the controlled journey in the Atlantic and the manipulated journeys of the Atlantic and Pacific

Source: Author

Most importantly, every journey exceeded the GFI base target (GFI BT 2028) by at least 49.32 gCO2/MJ, but it must be considered that $Energy_j = Energy_{total}$ in this research, leaving

uncertainty that this will be the true difference in gCO2/MJ. Nonetheless, it is possible that a lot of ship operators' costs will be exceedingly high, regardless of journey that a ship operator chooses to sail. Ship operators may want to consider intermodal transportation or switching to a near-zero fuel or fuel-less ship to decrease GFI and overall cost. This, however, does not seem practical.

This research can be helpful to ship operators because it can present a closer idea of what reality might be (with more, accurate data) once implementation begins with the IMO. This can also be beneficial to supply chain analysts of ship operating companies when they want to minimize costs from sailing, especially with the new IMO Net-Zero Regulation. All in all, when comparing the two journeys, most of the results were easy to compare and a two-sample t-test would determine if the average cost difference was significant or not.

Nonetheless, ship costs to sail from POL to POD will increase. Ship operators need to be prepared on how to remain competitive in multiple markets with the new regulations from the IMO. Using the results from this research, ship operators might investigate different ship engines for their ships, invest in cleaner fuel than MGO and VLSFO, ships might also slow-steam their ships, which was mentioned by Fagerholt et al. (2015), from POL to POD. These results can be a steppingstone for ship operators to prepare for a possible costly future.

5. Conclusion

In conclusion, the research was constructed to incorporate the new IMO regulation that will impact the future of shipping. From a supply chain standpoint, this will affect how a ship reaches its destination in the realm of international logistics. By establishing the existing regulations from the IMO such as ECAs, fuel emission standards (VLSFO and MGO) and learning how ships operate inside and outside of those areas, this research adds to the building of this knowledge by integrating penalties for inability to reach GFI 2028 base and direct compliance targets for Net-Zero emissions. An interactive model was formed to take suitable data that would aid in the hypothetical scenarios from the mainland United States to Hawaii and the US Caribbean. Maximum speed, average turnaround time, average operational cost, average power consumption, MGO and LFO consumption, and average load percentage were implemented into their respective equations or considered when analyzing the results. Interestingly enough, there was more of a wide range of differences between the costs of the controlled journey and the manipulated journey in all the hypothetical scenarios. However, the results show that there was no significant difference in the average cost to say that sailing the shortest route is more cost effective than incorporating the evasion route when sailing. Therefore, more information is needed to effectively establish the impact that the new IMO Net Zero regulation will have on the future of shipping.

Nonetheless, this research has displayed what the possible effects that this new regulation has on cost of shipping and the answer to the research question proposed in the beginning of this study is: a tentative yes, ship routing might be more inclined to change with the new IMO Net Zero regulation to reduce cost. That is, further research will be necessary in order to confidently confirm yes or no. This is because of the limitations when incorporating total energy into this study. Ship energy data to sufficiently classify total energy consumed by a ship is needed to fill the gaps in this research, if done correctly then the answer to the research question may become more conclusive and beneficial to the academia in international supply chains and logistics.

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